UNIVERSITY TEXTBOOK SERIES IN MATHEMATICAL SCIENCES

UNIVERSITY MATHEMATICS I

Analysis, Mathematical Logic & Algebraic Foundations

Comprehensive Theory, Formal Proofs & Worked Examples

Designed for First-Year Physics, Chemistry & Engineering Undergraduates

Covering Propositional Logic, Quantifiers, Set Theory, Binary Relations, Groups, Rings, The Real Completeness Axiom, Numerical Sequences, ε-δ Continuity, Differential Calculus, Rolle's & Mean Value Theorems, Taylor Expansions, and Riemann Integration.

Enriched with authentic historical portraits of pioneer mathematicians from Wikimedia Commons, precision TikZ analytical diagrams, rigorous proofs from scratch, and in-depth multi-step worked examples.

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Department of Sciences · First Edition

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Preface

This textbook provides a rigorous, exhaustive, and pedagogical presentation of **University Mathematics I** (Calculus, Real Analysis, Mathematical Logic, and Abstract Algebraic Structures). Designed specifically for undergraduate students in the physical sciences, chemistry, and engineering, it bridges the gap between high school calculus and axiomatic modern university mathematics.

Pedagogical Philosophy and Structure

1. **Axiomatic Rigor from First Principles:** Mathematical definitions are stated with uncompromising precision using standard first-order logical quantifiers ($\forall, \exists, \exists!$). Key theorems (Completeness of $\mathbb{R}$, Bolzano-Weierstrass, Monotone Convergence, Intermediate Value Theorem, Rolle, Mean Value Theorem, Taylor with Lagrange remainder, Fundamental Theorem of Calculus) are accompanied by complete, step-by-step proofs.

2. **Geometric and Analytical Visualizations:** Concepts are illuminated with standalone TikZ vector diagrams depicting set operations, mapping classifications, the supremum characterization, ε-N sequence convergence tubes, ε-δ continuity neighborhoods, tangent-secant Mean Value parallelism, and Riemann sum approximations.

3. **Historical Milestones and Human Dimension:** Authentic historical portraits of foundational thinkers (Boole, Cantor, Dedekind, Cauchy, Weierstrass, Bolzano, Newton, Leibniz, Euler, Taylor, Lagrange, Riemann, Galois, Rolle) provide context for the evolutionary milestones of mathematical thought.

4. **Multi-Step Worked Examples:** Each chapter concludes with detailed worked examples demonstrating how abstract theorems translate into concrete proof techniques and computational mastery.

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Chapter 1

Mathematical Logic, Proof Methods & Set Theory

1.1 Propositional Logic and Logical Connectives

Mathematical analysis is built upon the foundation of formal logic. A **mathematical assertion** (or proposition) is a declarative statement that is unambiguously either **True** (T or 1) or **False** (F or 0), but never both simultaneously (the *Law of Excluded Middle* and the *Principle of Non-Contradiction*).

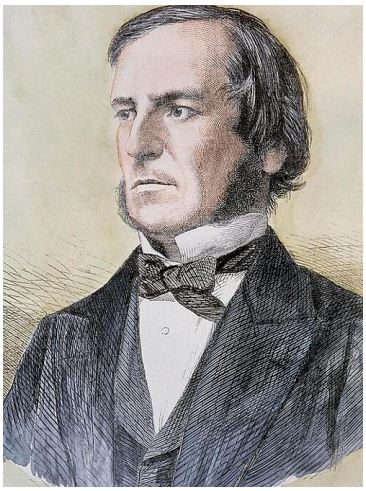


Figure 1.1: George Boole (1815–1864), English mathematician and philosopher who formalized symbolic mathematical logic in *The Laws of Thought* (1854), laying the algebraic foundation of modern computing and formal mathematics.

Definition · Elementary Logical Connectives

Let P and Q be two propositions. We define compound propositions via truth functions:

1. **Negation ($\neg P$ or $\overline{P}$):** "Not P". True if and only if P is false.
2. **Conjunction ($P \wedge Q$):** "P and Q". True if and only if both P and Q are simultaneously true.
3. **Disjunction ($P \vee Q$):** "P or Q" (inclusive or). True if at least one of P or Q is true; false only if both are false.
4. **Implication ($P \implies Q$):** "P implies Q" or "If P, then Q". False if and only if P is true and Q is false. By definition, $P \implies Q$ is logically equivalent to $(\neg P) \vee Q$.
5. **Equivalence ($P \iff Q$):** "P is equivalent to Q" or "P if and only if Q". True if both P and Q possess the same truth value. Logically equivalent to $(P \implies Q) \wedge (Q \implies P)$.

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \implies Q$	$P \iff Q$
T	T	F	T	T	T	T
T	F	F	F	T	F	F
F	T	T	F	T	T	F
F	F	T	F	F	T	T

Proposition / Lemma · Essential Tautologies and De Morgan's Logical Laws

For any propositions P, Q, R, the following compound propositions are **tautologies** (true under all truth assignments):

1. **Double Negation:** $\neg(\neg P) \iff P$.
2. **De Morgan's Laws:**

$$\neg(P \wedge Q) \iff (\neg P) \vee (\neg Q), \qquad \neg(P \vee Q) \iff (\neg P) \wedge (\neg Q).$$

3. **Distributivity:**

$$P \wedge (Q \vee R) \iff (P \wedge Q) \vee (P \wedge R), \qquad P \vee (Q \wedge R) \iff (P \vee Q) \wedge (P \vee R).$$

4. **Contrapositive Equivalence (Transposition):**

$$(P \implies Q) \iff (\neg Q \implies \neg P).$$

5. **Reductio ad Absurdum (Contradiction):**

$$(P \implies Q) \iff [(P \wedge \neg Q) \implies \text{Contradiction}].$$

1.2 Universal and Existential Quantifiers

In mathematical analysis, assertions typically contain variables evaluated over specified sets. A statement containing a variable $x \in E$ is called an **open sentence** or **predicate**, denoted $P(x)$.

Definition · Quantifiers and Their Scopes

1. **Universal Quantifier ($\forall$, "for all" / "for every"):**

$$\forall x \in E, \quad P(x)$$

is true if and only if $P(x)$ holds true for every single element x in the domain E.

2. **Existential Quantifier ($\exists$, "there exists at least one"):**

$$\exists x \in E, \quad P(x)$$

is true if and only if there is at least one element $x \in E$ for which $P(x)$ is true.

3. **Uniqueness Quantifier ($\exists!$, "there exists a unique"):**

$$\exists! x \in E, \quad P(x) \iff [(\exists x \in E, P(x)) \wedge (\forall x, y \in E, (P(x) \wedge P(y)) \implies x = y)].$$

Theorem · Negation of Quantified Statements

The negation of quantified statements swaps the quantifiers and negates the inner predicate:

$$\neg\left(\forall x \in E,\ P(x)\right) \iff \exists x \in E,\ \neg P(x), \tag{1.1}$$
$$\neg\left(\exists x \in E,\ P(x)\right) \iff \forall x \in E,\ \neg P(x). \tag{1.2}$$

For multi-quantifier propositions, the order of quantifiers is critical and cannot be arbitrarily exchanged:

$$\exists y \in F,\ \forall x \in E,\ P(x,y) \implies \forall x \in E,\ \exists y \in F,\ P(x,y),$$

but the converse is **false** in general!

1.3 Methods of Mathematical Proof

1. **Direct Proof:** Assume P is true. Through a sequence of rigorous deductions using axioms, definitions, and established theorems, show that Q necessarily follows.
2. **Proof by Contrapositive:** To prove $P \implies Q$, we prove the logically equivalent contrapositive statement $\neg Q \implies \neg P$. This is especially powerful when $\neg Q$ provides more concrete algebraic structure than P.
3. **Proof by Contradiction (*Reductio ad Absurdum*):** To prove an assertion A, assume its negation $\neg A$ is true. Derive a logical contradiction (such as $0 = 1$ or $x \in B \wedge x \notin B$). Therefore, $\neg A$ must be false, establishing that A is true.
4. **Disproof by Counterexample:** To disprove a universal assertion $\forall x \in E, P(x)$, it suffices to produce a single concrete element $x_0 \in E$ for which $P(x_0)$ is false.
5. **Proof by Mathematical Induction:**

Theorem · Principle of Mathematical Induction (Peano Axiom)

Let $P(n)$ be a predicate defined for all integers $n \geq n_0$ ($n_0 \in \mathbb{Z}$). If:

(a) **Base Step:** $P(n_0)$ is true;

(b) **Inductive Step:** For any arbitrary integer $k \geq n_0$, the assumption that $P(k)$ is true (the *induction hypothesis*) implies that $P(k+1)$ is true:

$$\forall k \geq n_0, \quad P(k) \implies P(k+1);$$

Then $P(n)$ is true for all integers $n \geq n_0$.

1.4 Axiomatic Set Theory and Operations

A **set** is an intuitive collection of well-defined, distinct mathematical objects called its **elements**. If an object x belongs to a set A, we write $x \in A$; otherwise $x \notin A$.

Definition · Subsets, Equality, and the Power Set

Let A and B be sets in a universe E:

1. **Inclusion (Subset):** $A \subseteq B \iff \forall x, (x \in A \implies x \in B)$.
2. **Set Equality (Double Inclusion):** $A = B \iff (A \subseteq B \wedge B \subseteq A)$.

3. **Empty Set ($\emptyset$):** The unique set containing no elements. For every set A, $\emptyset \subseteq A$.
4. **Power Set ($\mathcal{P}(E)$):** The set of all subsets of E:

$$\mathcal{P}(E) = \{A \mid A \subseteq E\}.$$

If E is finite with $|E| = n$ elements, its power set has $|\mathcal{P}(E)| = 2^n$ elements.

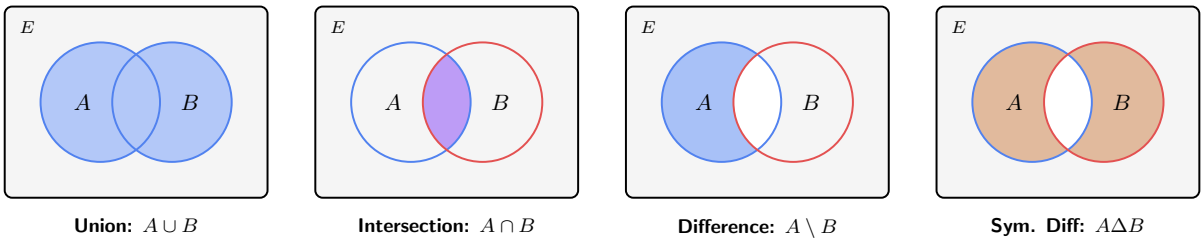


Figure 1.2: Fundamental Set Operations: Geometric Venn diagram representation of Union ($A \cup B$), Intersection ($A \cap B$), Difference ($A \setminus B$), and Symmetric Difference ($A \Delta B$).

Definition · Fundamental Set Operations

Let $A, B \in \mathcal{P}(E)$ be subsets of an ambient set E:

1. **Union:** $A \cup B = \{x \in E \mid x \in A \vee x \in B\}$.
2. **Intersection:** $A \cap B = \{x \in E \mid x \in A \wedge x \in B\}$. If $A \cap B = \emptyset$, A and B are **disjoint**.
3. **Set Difference (Relative Complement):** $A \setminus B = \{x \in E \mid x \in A \wedge x \notin B\}$.
4. **Complement in E:** $\complement_E A = E \setminus A = \{x \in E \mid x \notin A\}$.
5. **Symmetric Difference:** $A \Delta B = (A \setminus B) \cup (B \setminus A) = (A \cup B) \setminus (A \cap B)$.
6. **Cartesian Product:** $A \times B = \{(a, b) \mid a \in A \wedge b \in B\}$.

Theorem · De Morgan's Laws for Sets

For any subsets $A, B \subseteq E$:

$$\complement_E(A \cup B) = \complement_E A \cap \complement_E B, \qquad \complement_E(A \cap B) = \complement_E A \cup \complement_E B. \tag{1.3}$$

More generally, for any indexed family of subsets $\{A_i\}_{i \in I}$:

$$\complement_E\left(\bigcup_{i \in I} A_i\right) = \bigcap_{i \in I} \complement_E A_i, \qquad \complement_E\left(\bigcap_{i \in I} A_i\right) = \bigcup_{i \in I} \complement_E A_i. \tag{1.4}$$

1.5 Formal Syntax of Sentential Logic, Unique Readability & Functional Completeness

To study mathematical logic with mathematical rigor, the language of logic itself must be formalized as a mathematical object. This foundational perspective distinguishes **syntax** (pure string formation and grammar) from **semantics** (truth values and interpretations).

1.5.1 Strings, Alphabets, and Well-Formed Formulas

Definition · The Formal Language of Sentential Logic $\mathcal{L}_S$

The formal alphabet of sentential logic consists of three disjoint sets of symbols:

1. **Sentence Symbols:** A countably infinite set $\mathcal{S} = \{A_1, A_2, A_3, \dots\} \cup \{P, Q, R, \dots\}$.
2. **Logical Connectives:** The unary symbol $\neg$ (negation) and binary symbols $\{\wedge, \vee, \rightarrow, \leftrightarrow\}$.
3. **Punctuation Symbols:** Left and right parentheses (and).

An **expression** is any finite sequence (string) of symbols from this alphabet. The concatenation of two expressions α and β is denoted $\alpha\beta$.

Not every string represents a syntactically valid statement (e.g., $) \rightarrow P($ is meaningless). We define the set of **well-formed formulas** (WFF) inductively from first principles.

Definition · Inductive Definition of Well-Formed Formulas (WFF)

The set WFF of well-formed sentential formulas is the smallest set of strings satisfying:

1. Every sentence symbol $A \in \mathcal{S}$ is in WFF (atomic formulas).
2. If $\alpha \in$ WFF, then the string $(\neg\alpha) \in$ WFF.
3. If $\alpha, \beta \in$ WFF and $\circ \in \{\wedge, \vee, \rightarrow, \leftrightarrow\}$, then the string $(\alpha \circ \beta) \in$ WFF.
4. No string is in WFF unless its membership follows from a finite number of applications of rules (1)–(3).

1.5.2 The Unique Readability Theorem

In natural language, syntactic ambiguity often creates paradoxical interpretations (e.g., "time flies like an arrow"). In formal logic, mathematical reasoning requires that every compound formula possesses an **unambiguous parse tree**: its principal connective and component subformulas must be uniquely determined.

Proposition / Lemma · Parenthesis Balance Lemma for Well-Formed Formulas

For any string w, let $L(w)$ and $R(w)$ denote the total number of left and right parentheses in w, respectively. For every $\alpha \in$ WFF:

1. $L(\alpha) = R(\alpha)$ (every well-formed formula contains an equal number of left and right parentheses).
2. For every **non-empty proper initial segment** (prefix) u of α, $L(u) > R(u)$.

Proof. We proceed by structural induction on the construction of $\alpha \in$ WFF.

- *Base Case:* If $\alpha = A \in \mathcal{S}$ is an atomic symbol, $L(\alpha) = R(\alpha) = 0$. Since α has length 1, its only non-empty proper prefix is the empty string or non-existent, satisfying both claims vacuously.
- *Negation Step:* Let $\alpha = (\neg\beta)$ where $\beta \in$ WFF satisfies the inductive hypothesis. Then $L(\alpha) = 1 + L(\beta) = 1 + R(\beta) = R(\alpha)$. Any non-empty proper prefix u of α is either: (i) (, where $L(u) = 1 > R(u) = 0$; (ii) $(\neg$, where $L(u) = 1 > R(u) = 0$; (iii) $(\neg u'$ where u' is a prefix of β, giving $L(u) = 1 + L(u') \geq 1 + R(u') > R(u') = R(u)$; (iv) $(\neg\beta$, giving $L(u) = 1 + L(\beta) = 1 + R(\beta) > R(\beta) = R(u)$. In every case, $L(u) > R(u)$.

- *Binary Connective Step:* Let $\alpha = (\beta \circ \gamma)$ where $\beta, \gamma \in \text{WFF}$. Then $L(\alpha) = 1 + L(\beta) + L(\gamma) = 1 + R(\beta) + R(\gamma) = R(\alpha)$. Any proper prefix u intersects $(\beta$, or $(\beta\circ$, or $(\beta \circ \gamma'$ with γ' a prefix of γ. In all subcases, the initial unmatched left parenthesis guarantees $L(u) > R(u)$. ■

Theorem · Unique Readability of Sentential Formulas

For any formula $\alpha \in \text{WFF}$:

1. No proper initial segment of α belongs to WFF.
2. Exactly one of the following mutually exclusive conditions holds:
   - $\alpha \in \mathcal{S}$ is an atomic sentence symbol.
   - There is a unique formula $\beta \in \text{WFF}$ such that $\alpha = (\neg\beta)$.
   - There exist unique formulas $\beta, \gamma \in \text{WFF}$ and a unique binary connective $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$ such that $\alpha = (\beta \circ \gamma)$.

1.5.3 Truth Assignments and Functional Completeness

Definition · Truth Valuations and Homomorphic Extension

A **truth assignment** is a function $v : \mathcal{S} \to \{0, 1\}$. By the Unique Readability Theorem, v extends uniquely to a valuation function $\bar{v} : \text{WFF} \to \{0, 1\}$ satisfying:

$$\begin{aligned}
\bar{v}(A) &= v(A) \quad \text{for all } A \in \mathcal{S}, \\
\bar{v}((\neg\alpha)) &= 1 - \bar{v}(\alpha), \\
\bar{v}((\alpha \wedge \beta)) &= \min(\bar{v}(\alpha), \bar{v}(\beta)) = \bar{v}(\alpha) \cdot \bar{v}(\beta), \\
\bar{v}((\alpha \vee \beta)) &= \max(\bar{v}(\alpha), \bar{v}(\beta)) = \bar{v}(\alpha) + \bar{v}(\beta) - \bar{v}(\alpha)\bar{v}(\beta), \\
\bar{v}((\alpha \to \beta)) &= \begin{cases} 0 & \text{if } \bar{v}(\alpha) = 1 \text{ and } \bar{v}(\beta) = 0, \\ 1 & \text{otherwise.} \end{cases}
\end{aligned}$$

Definition · Boolean Functions and Functional Completeness

An n-ary **Boolean function** is a map $F : \{0, 1\}^n \to \{0, 1\}$. There are exactly 2^{2^n} distinct n-ary Boolean functions. A set of logical connectives Ω is **functionally complete** (or adequate) if every Boolean function F can be expressed by a formula using only connectives from Ω.

Theorem · Normal Form Theorems and Complete Connective Bases

1. **Disjunctive Normal Form (DNF):** Every Boolean function $F(x_1, \ldots, x_n)$ can be written as a disjunction of minterms:

$$F(x_1, \ldots, x_n) = \bigvee_{(c_1,\ldots,c_n) \in F^{-1}(1)} \left(\ell_1^{c_1} \wedge \ell_2^{c_2} \wedge \cdots \wedge \ell_n^{c_n}\right), \quad \text{where } \ell_i^1 = x_i \text{ and } \ell_i^0 = \neg x_i.$$

   Consequently, the basis $\{\neg, \wedge, \vee\}$ is functionally complete.

2. **Minimal Binary Bases:** Since $P \vee Q \iff \neg(\neg P \wedge \neg Q)$ and $P \wedge Q \iff \neg(\neg P \vee \neg Q)$, both $\{\neg, \wedge\}$ and $\{\neg, \vee\}$ are functionally complete.

3. **Singleton Bases (Sheffer Stroke and Peirce Arrow):** The NAND operator ($P \uparrow Q \iff \neg(P \wedge Q)$) and NOR operator ($P \downarrow Q \iff \neg(P \vee Q)$) are each individually functionally

complete:

$$\neg P \iff P \uparrow P, \qquad P \wedge Q \iff (P \uparrow Q) \uparrow (P \uparrow Q), \qquad P \vee Q \iff (P \uparrow P) \uparrow (Q \uparrow Q).$$

Proposition / Lemma · Incompleteness of $\{\wedge, \vee\}$ (Monotonicity Barrier)

The set of connectives $\{\wedge, \vee\}$ is **not** functionally complete. In particular, it cannot express negation $\neg P$.

Proof. Every formula built solely from $\{\wedge, \vee\}$ defines a **monotone Boolean function**: if $v(A) = 1$ for all sentence symbols, then $\bar{v}(\alpha) = 1$. Since $\neg P$ evaluates to 0 when $P = 1$, negation cannot be monotone and therefore cannot be expressed using $\{\wedge, \vee\}$ alone. ■

1.6 First-Order Languages, Structures & Model Theory

Sentential logic cannot formalize internal mathematical relations such as "$x < y$" or arithmetic operations such as "$x + y = z$". **First-Order Logic (Predicate Logic)** provides the foundational framework for modern axiomatic mathematics.

1.6.1 Syntax of First-Order Logic

Definition · First-Order Signatures and Expressions

A **first-order signature** (or language) $\mathcal{L}$ is defined by three disjoint sets of non-logical symbols:

1. A set $\mathcal{C}$ of **constant symbols** (e.g., $0, 1, e, \pi$).
2. A set $\mathcal{F}$ of **function symbols**, each assigned an integer arity $n \geq 1$ (e.g., $+$, $\cdot$, exp).
3. A set $\mathcal{R}$ of **relation (predicate) symbols**, each assigned an integer arity $n \geq 1$ (e.g., $<$, $\leq$, $\in$).

The logical symbols consist of variables $V = \{v_0, v_1, v_2, \dots\}$, connectives $\{\neg, \to\}$, universal quantifier $\forall$, equality symbol $=$, and parentheses.

Definition · Terms and Formulas

1. **Terms (Term($\mathcal{L}$)):** The formal "nouns" denoting objects.
   - Every variable $x \in V$ and constant $c \in \mathcal{C}$ is a term.
   - If $f \in \mathcal{F}$ has arity n and $t_1, \dots, t_n$ are terms, then $f(t_1, \dots, t_n)$ is a term.
2. **Atomic Formulas:** If t_1, t_2 are terms and $R \in \mathcal{R}$ has arity n, then:
   $$t_1 = t_2 \quad \text{and} \quad R(t_1, \dots, t_n) \quad \text{are atomic formulas.}$$
3. **Well-Formed Formulas (WFF($\mathcal{L}$)):**
   - Every atomic formula is in WFF($\mathcal{L}$).
   - If $\varphi, \psi \in \text{WFF}(\mathcal{L})$ and $x \in V$, then $(\neg\varphi)$, $(\varphi \to \psi)$, and $(\forall x \varphi)$ are in WFF($\mathcal{L}$).
   - Standard abbreviations: $\exists x \varphi \equiv \neg\forall x(\neg\varphi)$, $\varphi \wedge \psi \equiv \neg(\varphi \to \neg\psi)$, $\varphi \vee \psi \equiv (\neg\varphi) \to \psi$.

Definition · Free Variables and Sentences

The set of **free variables** $\mathrm{Free}(\varphi)$ of a formula φ is defined recursively:

$$\begin{aligned}
\mathrm{Free}(t_1 = t_2) &= \mathrm{Var}(t_1) \cup \mathrm{Var}(t_2), \\
\mathrm{Free}(R(t_1, \ldots, t_n)) &= \bigcup_{i=1}^{n} \mathrm{Var}(t_i), \\
\mathrm{Free}(\neg\varphi) &= \mathrm{Free}(\varphi), \\
\mathrm{Free}(\varphi \to \psi) &= \mathrm{Free}(\varphi) \cup \mathrm{Free}(\psi), \\
\mathrm{Free}(\forall x \varphi) &= \mathrm{Free}(\varphi) \setminus \{x\}.
\end{aligned}$$

A formula σ with $\mathrm{Free}(\sigma) = \emptyset$ is called a **sentence** (or closed formula).

1.6.2 Structures and Tarskian Semantics

Definition · First-Order Structures

An **$\mathcal{L}$-structure** $\mathcal{M}$ is a pair (M, I) where:

1. M is a non-empty set called the **universe** (or domain) of $\mathcal{M}$, denoted $|\mathcal{M}|$.
2. The interpretation function I assigns:
   - For each constant $c \in \mathcal{C}$, an element $c^{\mathcal{M}} = I(c) \in M$.
   - For each n-ary function symbol $f \in \mathcal{F}$, a function $f^{\mathcal{M}} : M^n \to M$.
   - For each n-ary relation symbol $R \in \mathcal{R}$, an n-ary relation $R^{\mathcal{M}} \subseteq M^n$.

Definition · Tarskian Satisfaction Relation $\mathcal{M} \models \varphi[s]$

Let $\mathcal{M}$ be an $\mathcal{L}$-structure and $s : V \to M$ a **variable assignment**. The valuation of a term t, denoted $\bar{s}(t) \in M$, evaluates variables via $s(x)$ and functions via $f^{\mathcal{M}}$. The **satisfaction relation** $\mathcal{M} \models \varphi[s]$ is defined inductively:

$$\begin{aligned}
\mathcal{M} \models (t_1 = t_2)[s] &\iff \bar{s}(t_1) = \bar{s}(t_2) \text{ in } M, \\
\mathcal{M} \models R(t_1, \ldots, t_n)[s] &\iff (\bar{s}(t_1), \ldots, \bar{s}(t_n)) \in R^{\mathcal{M}}, \\
\mathcal{M} \models (\neg\varphi)[s] &\iff \mathcal{M} \not\models \varphi[s], \\
\mathcal{M} \models (\varphi \to \psi)[s] &\iff (\mathcal{M} \models \varphi[s] \implies \mathcal{M} \models \psi[s]), \\
\mathcal{M} \models (\forall x \varphi)[s] &\iff \forall a \in M,\ \mathcal{M} \models \varphi[s(x \mapsto a)],
\end{aligned}$$

where $s(x \mapsto a)$ denotes the assignment identical to s except that x is mapped to a. For a sentence σ, $\mathcal{M} \models \sigma[s]$ is independent of s; we write $\mathcal{M} \models \sigma$ ("$\mathcal{M}$ is a model of σ").

1.6.3 Definability and Automorphisms

Definition · Definable Subsets and Relations

Let $\mathcal{M}$ be an $\mathcal{L}$-structure. A subset $S \subseteq M^k$ is **definable without parameters** if there exists a formula $\varphi(x_1, \ldots, x_k)$ such that:

$$S = \left\{ (a_1, \ldots, a_k) \in M^k \;\middle|\; \mathcal{M} \models \varphi[a_1, \ldots, a_k] \right\}.$$

Theorem · Automorphism Invariance and Non-Definability Criterion

An **automorphism** of $\mathcal{M}$ is a bijection $\pi : M \to M$ that preserves all constants, functions, and relations:

$$\pi(c^{\mathcal{M}}) = c^{\mathcal{M}}, \quad \pi(f^{\mathcal{M}}(a_1, \dots)) = f^{\mathcal{M}}(\pi(a_1), \dots), \quad (a_1, \dots) \in R^{\mathcal{M}} \iff (\pi(a_1), \dots) \in R^{\mathcal{M}}.$$

1. **Invariance Theorem:** If $\varphi(x_1, \dots, x_k)$ is any first-order formula and $\pi \in \mathrm{Aut}(\mathcal{M})$, then for all $a_1, \dots, a_k \in M$:

$$\mathcal{M} \models \varphi[a_1, \dots, a_k] \iff \mathcal{M} \models \varphi[\pi(a_1), \dots, \pi(a_k)].$$

2. **Non-Definability Criterion:** If $S \subseteq M^k$ is definable without parameters, then $\pi(S) = S$ for every $\pi \in \mathrm{Aut}(\mathcal{M})$. Consequently, if an automorphism π maps an element of S outside of S, then S is **not first-order definable**.

1.6.4 Elementary Equivalence and Elementary Substructures

Definition · Elementary Classes and Elementary Equivalence

1. Two structures $\mathcal{M}$ and $\mathcal{N}$ are **elementarily equivalent** (denoted $\mathcal{M} \equiv \mathcal{N}$) if for every first-order sentence σ:

$$\mathcal{M} \models \sigma \iff \mathcal{N} \models \sigma.$$

2. Isomorphic structures are always elementarily equivalent: $\mathcal{M} \cong \mathcal{N} \implies \mathcal{M} \equiv \mathcal{N}$. However, the converse fails: the ordered fields $(\mathbb{Q}, <)$ and $(\mathbb{R}, <)$ are dense linear orders without endpoints, so $(\mathbb{Q}, <) \equiv (\mathbb{R}, <)$, yet they cannot be isomorphic because $|\mathbb{Q}| = \aleph_0 \neq 2^{\aleph_0} = |\mathbb{R}|$.
3. A class $\mathcal{K}$ of $\mathcal{L}$-structures is **elementary** if there is a set of sentences Σ such that $\mathcal{K} = \mathrm{Mod}(\Sigma) = \{\mathcal{M} \mid \mathcal{M} \models \Sigma\}$. If Σ can be chosen finite (or a single sentence), $\mathcal{K}$ is **strongly elementary**.

Definition · Substructures and Elementary Substructures

Let $\mathcal{A}$ and $\mathcal{B}$ be $\mathcal{L}$-structures with $A \subseteq B$.

1. $\mathcal{A}$ is a **substructure** of $\mathcal{B}$ (denoted $\mathcal{A} \subseteq \mathcal{B}$) if the inclusion map $j : A \hookrightarrow B$ is an embedding.
2. $\mathcal{A}$ is an **elementary substructure** of $\mathcal{B}$ (denoted $\mathcal{A} \preceq \mathcal{B}$) if for every formula $\varphi(x_1, \dots, x_n)$ and all $a_1, \dots, a_n \in A$:

$$\mathcal{A} \models \varphi[a_1, \dots, a_n] \iff \mathcal{B} \models \varphi[a_1, \dots, a_n].$$

3. **Tarski-Vaught Test:** $\mathcal{A} \preceq \mathcal{B} \iff$ for every formula $\varphi(x, y_1, \dots, y_n)$ and $a_1, \dots, a_n \in A$, if $\mathcal{B} \models \exists x \varphi(x, a_1, \dots, a_n)$, then there exists an element $c \in A$ such that $\mathcal{B} \models \varphi(c, a_1, \dots, a_n)$.

1.7 The Compactness Theorem & Non-Axiomatizability

The Compactness Theorem is the cornerstone of Model Theory. It connects finite local satisfiability to global infinite existence, providing a powerful tool for discovering non-standard structures and proving non-axiomatizability results.

1.7.1 Compactness in Sentential Logic: Topological and Combinatorial Proofs

Theorem · Sentential Compactness Theorem

A set of sentential formulas Σ is satisfiable if and only if every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable.

Topological Proof via the Cantor Space $2^{\mathbb{N}}$. Let $\mathcal{S} = \{A_1, A_2, \dots\}$ be the countable set of sentence symbols. A truth assignment is a sequence $v = (v(A_1), v(A_2), \dots) \in \{0,1\}^{\mathbb{N}}$. Endow the space $X = \{0,1\}^{\mathbb{N}}$ with the product topology (the Cantor Space).

1. By **Tychonoff's Theorem** (or the Heine-Borel theorem for Cantor sets), X is a **compact topological space**.
2. For each formula $\alpha \in \text{WFF}$, the set of satisfying assignments $[\alpha] = \{v \in X \mid \bar{v}(\alpha) = 1\}$ depends only on the finitely many sentence symbols occurring in α. Therefore, $[\alpha]$ is a finite union of basic cylinder sets, meaning $[\alpha]$ is both **closed and open** (clopen) in X.
3. Suppose every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable. Then the finite intersection $\bigcap_{\alpha \in \Sigma_0} [\alpha] \neq \emptyset$. Thus, the family of closed sets $\{[\alpha]\}_{\alpha \in \Sigma}$ possesses the **Finite Intersection Property (FIP)**.
4. Since X is compact, the total intersection is non-empty:

$$\bigcap_{\alpha \in \Sigma} [\alpha] \neq \emptyset.$$

Any assignment v^* in this intersection satisfies every formula in Σ simultaneously. ■

1.7.2 Application: The de Bruijn–Erdős Graph Coloring Theorem

Theorem · The de Bruijn–Erdős Theorem (1951)

Let $G = (V, E)$ be an arbitrary (possibly uncountably infinite) graph, and let $k \in \mathbb{N}_{\geq 1}$. The graph G is k-colorable if and only if every finite subgraph $G_0 \subseteq G$ is k-colorable.

Proof via Sentential Compactness. Define a propositional language with sentence symbols $\{P_{v,c} \mid v \in V,\, c \in \{1, 2, \dots, k\}\}$, where $P_{v,c}$ asserts that "vertex v is assigned color c". Construct the theory Σ with three families of axioms:

1. **At least one color per vertex:** For every $v \in V$: $\bigvee_{c=1}^{k} P_{v,c}$.
2. **At most one color per vertex:** For every $v \in V$ and distinct $c \neq c'$: $\neg(P_{v,c} \wedge P_{v,c'})$.
3. **Proper coloring across edges:** For every edge $\{u, v\} \in E$ and each color c: $\neg(P_{u,c} \wedge P_{v,c})$.

Now let $\Sigma_0 \subseteq \Sigma$ be an arbitrary finite subset. Σ_0 mentions only a finite set of vertices $V_0 \subset V$. The induced subgraph $G[V_0]$ is finite, and by hypothesis, it admits a valid k-coloring. Setting $v(P_{v,c}) = 1$ if vertex v receives color c defines a valuation satisfying Σ_0. By the Sentential Compactness Theorem, Σ is globally satisfiable, producing a valid k-coloring for the entire graph G. ■

1.7.3 The First-Order Compactness Theorem

Theorem · First-Order Compactness Theorem

1. **Model Form:** A set of first-order sentences Σ has a model if and only if every finite subset $\Sigma_0 \subseteq \Sigma$ has a model.
2. **Implication Form:** If $\Sigma \models \varphi$, then there exists a finite subset $\Sigma_0 \subseteq \Sigma$ such that $\Sigma_0 \models \varphi$.

1.7.4 Proving Non-Axiomatizability via Compactness

Proposition / Lemma · Non-Standard Models of Arithmetic

There exists a non-standard model of True Arithmetic: an ordered structure $\mathcal{M}^* \equiv (\mathbb{N}, +, \cdot, 0, 1, <)$ that contains an "infinite" number c strictly greater than every standard integer ($c > n$ for all $n \in \mathbb{N}$).

Proof. Expand the language of arithmetic by adding a new constant symbol c. Consider the theory:

$$\Sigma = \mathrm{Th}(\mathbb{N}, +, \cdot, 0, 1, <) \cup \{c > \bar{n} \mid n \in \mathbb{N}\},$$

where $\bar{n}$ is the numeral $\underbrace{1 + 1 + \cdots + 1}_{n \text{ times}}$. Let $\Sigma_0 \subset \Sigma$ be any finite subset. Σ_0 contains axioms from $\mathrm{Th}(\mathbb{N})$ and finitely many conditions $c > \bar{n}_1, \ldots, c > \bar{n}_k$. Let $N = \max(n_1, \ldots, n_k) + 1$. In the standard structure $\mathbb{N}$, interpreting c as N satisfies all sentences in Σ_0. By Compactness, Σ has a model $\mathcal{M}^*$. In $\mathcal{M}^*$, all first-order theorems of arithmetic remain true, yet $c^{\mathcal{M}^*}$ is greater than every standard integer! ■

Proposition / Lemma · Non-Axiomatizability of Finiteness and Graph Connectivity

1. **Finiteness is Not Elementary:** If a theory Σ has arbitrarily large finite models, then it possesses an infinite model. Consequently, the class of finite groups, finite graphs, or finite fields is **not** first-order axiomatizable.
2. **Graph Connectivity is Not Elementary:** The class of connected graphs cannot be axiomatized in first-order logic.
3. **Fields of Characteristic 0:** The class Field_0 is elementary (axiomatized by the infinite set $\{\bar{p} \neq 0 \mid p \text{ prime}\}$), but **not strongly elementary**: no single sentence (or finite set of sentences) can define characteristic 0.

1.8 Formal Deductive Systems, Soundness & Gödel's Completeness Theorem

The semantic notion of truth ($\Sigma \models \varphi$) requires surveying all conceivable mathematical structures. In contrast, the syntactic notion of proof ($\Sigma \vdash \varphi$) operates purely via mechanical, algorithmic symbol manipulation. Kurt Gödel's landmark 1929 Completeness Theorem establishes that these two distinct concepts are in perfect harmony.



Figure 1.3: Richard Dedekind (1831–1916), who pioneered the axiomatic isolation of arithmetic and set-theoretic foundations, setting the stage for formal proof theory and Gödel's completeness theorem.

1.8.1 Hilbert-Style Formal Deductive System

Definition · Logical Axiom Schemata and Rules of Inference

A formal deduction in first-order logic utilizes the following logical axioms:

1. **Group 1 (Tautologies):** All substitution instances of propositional tautologies.
2. **Group 2 (Quantifier Instantiation):** $\forall x\varphi \to \varphi_t^x$, where t is substitutable for x in φ.
3. **Group 3 (Quantifier Distribution):** $\forall x(\varphi \to \psi) \to (\forall x\varphi \to \forall x\psi)$.
4. **Group 4 (Vacuous Quantification):** $\varphi \to \forall x\varphi$, where $x \notin \text{Free}(\varphi)$.
5. **Group 5 (Equality Axioms):** Reflexivity ($x = x$), symmetry ($x = y \to y = x$), transitivity ($x = y \wedge y = z \to x = z$), and congruence for functions and relations.

The system possesses two **rules of inference**:

- **Modus Ponens (MP):** From φ and $\varphi \to \psi$, infer ψ.
- **Generalization (GEN):** From φ, infer $\forall x\varphi$.

A formula φ is **deducible** from Σ (denoted $\Sigma \vdash \varphi$) if there is a finite sequence of formulas $\psi_1, \ldots, \psi_n = \varphi$ where each ψ_i is an axiom, a member of Σ, or follows from earlier formulas via MP or GEN.

Theorem · The Deduction Metatheorem

Let Σ be a set of sentences and φ, ψ formulas.

$$\Sigma \cup \{\varphi\} \vdash \psi \iff \Sigma \vdash \varphi \to \psi.$$

Theorem · The Soundness Theorem

If $\Sigma \vdash \varphi$, then $\Sigma \models \varphi$.

Proof. By structural induction on the length n of the formal derivation. Every logical axiom is true in all structures under all assignments (validity). Modus Ponens preserves truth, and Generalization preserves validity. Therefore, every theorem is a semantic logical consequence. ■

1.8.2 The Henkin Model Construction and Gödel's Completeness Theorem

Definition · Consistency and Maximal Consistent Theories

1. A set of formulas Σ is **consistent** if $\Sigma \nvdash \bot$ (no contradiction can be formally derived).
2. **Lindenbaum's Lemma:** Every consistent theory Σ can be extended to a **maximal consistent theory** Σ^*, such that for every sentence φ, exactly one of $\varphi \in \Sigma^*$ or $\neg\varphi \in \Sigma^*$ holds.
3. **Henkin Witnessing Property:** A theory Σ has **witnesses** if for every formula $\varphi(x)$, there is a constant symbol c such that:

$$\exists x \varphi(x) \to \varphi(c) \in \Sigma.$$

Theorem · The Canonical Term Model Construction

Let Σ be a maximal consistent theory with witnesses in language $\mathcal{L}$. We construct the **Canonical Term Model** $\mathcal{M}$:

1. Define an equivalence relation on the set of closed terms $\mathrm{Term}_0(\mathcal{L})$:

$$t_1 \sim t_2 \iff \Sigma \vdash t_1 = t_2.$$

2. The universe is the quotient $M = \mathrm{Term}_0(\mathcal{L})/\sim$.
3. Constants: $c^{\mathcal{M}} = [c]$.
4. Functions: $f^{\mathcal{M}}([t_1], \ldots, [t_n]) = [f(t_1, \ldots, t_n)]$.
5. Relations: $([t_1], \ldots, [t_n]) \in R^{\mathcal{M}} \iff \Sigma \vdash R(t_1, \ldots, t_n)$.

Theorem · The Fundamental Truth Lemma and Gödel's Completeness Theorem

1. **The Truth Lemma:** For every sentence φ in the language of the term model:

$$\mathcal{M} \models \varphi \iff \varphi \in \Sigma.$$

   (Proved by structural induction on φ; the existential case is guaranteed by Henkin witnesses).
2. **Model Existence Theorem:** Every consistent first-order theory has a model.
3. **Gödel's Completeness Theorem (1929):** For any set of sentences Σ and sentence φ:

$$\boldsymbol{\Sigma \models \varphi \iff \Sigma \vdash \varphi}.$$

Proposition / Lemma · Compactness as a Corollary of Completeness

The First-Order Compactness Theorem follows immediately from Gödel's Completeness Theorem: If Σ is unsatisfiable, then $\Sigma \models \bot$. By Completeness, there is a formal deduction $\Sigma \vdash \bot$. Since a formal deduction is a **finite sequence** of formulas, it can use only finitely many premises $\Sigma_0 \subset \Sigma$. Thus $\Sigma_0 \vdash \bot$, which by Soundness implies $\Sigma_0 \models \bot$ (Σ_0 is unsatisfiable). Taking the contrapositive: if every finite subset $\Sigma_0 \subseteq \Sigma$ is satisfiable, then Σ is satisfiable!

1.9 Advanced Worked Examples with Step-by-Step Proofs

Worked Example 1.1 · Structural Induction: Sentence Symbols vs. Binary Connectives

Let $\alpha \in \text{WFF}$ be a well-formed sentential formula built strictly from sentence symbols and binary connectives $\{\wedge, \vee, \to, \leftrightarrow\}$. Let $S(\alpha)$ be the total number of occurrences of sentence symbols in α, and let $C(\alpha)$ be the total number of binary connectives in α. Prove by structural induction that:

$$S(\alpha) = C(\alpha) + 1.$$

Solution and Proof. We proceed by induction on the complexity of $\alpha \in \text{WFF}$.
1. Base Case: Let $\alpha = A \in \mathcal{S}$ be an atomic sentence symbol. Then $S(A) = 1$ and $C(A) = 0$. Thus $S(A) = 1 = 0 + 1 = C(A) + 1$, verifying the base case.
2. Inductive Step: Suppose α is a composite formula built with a binary connective: $\alpha = (\beta \circ \gamma)$ where $\circ \in \{\wedge, \vee, \to, \leftrightarrow\}$ and $\beta, \gamma \in \text{WFF}$. By the Inductive Hypothesis, the claim holds for both components:

$$S(\beta) = C(\beta) + 1 \quad \text{and} \quad S(\gamma) = C(\gamma) + 1.$$

By the syntactic structure of $\alpha = (\beta \circ \gamma)$:

- The sentence symbols in α are precisely the symbols in β plus the symbols in γ:

$$S(\alpha) = S(\beta) + S(\gamma).$$

- The binary connectives in α consist of those in β, those in γ, plus the new principal connective $\circ$:

$$C(\alpha) = C(\beta) + C(\gamma) + 1.$$

Substituting the induction hypotheses:

$$S(\alpha) = (C(\beta) + 1) + (C(\gamma) + 1) = [C(\beta) + C(\gamma) + 1] + 1 = C(\alpha) + 1.$$

By structural induction, $S(\alpha) = C(\alpha) + 1$ holds for all formulas. ■

Worked Example 1.2 · Functional Completeness: Expressing Connectives via the Sheffer Stroke

The Sheffer Stroke (NAND), denoted $\uparrow$, is defined by the truth condition:

$$P \uparrow Q \iff \neg(P \wedge Q).$$

Prove constructively that $\{\uparrow\}$ forms a functionally complete set of connectives by expressing $\neg P$, $P \wedge Q$, and $P \vee Q$ using only $\uparrow$.

Solution and Proof. Recall that the set $\{\neg, \wedge, \vee\}$ is functionally complete. It suffices to show that each can be synthesized from $\uparrow$:

1. **Negation ($\neg P$):** By definition:

$$P \uparrow P \iff \neg(P \wedge P) \iff \neg P.$$

   Hence, $\neg \mathbf{P} \iff \mathbf{P} \uparrow \mathbf{P}$.

2. **Conjunction ($P \wedge Q$):** Observe that $P \wedge Q \iff \neg(\neg(P \wedge Q)) \iff \neg(P \uparrow Q)$. Using our expression for negation:

$$\mathbf{P} \wedge \mathbf{Q} \iff (\mathbf{P} \uparrow \mathbf{Q}) \uparrow (\mathbf{P} \uparrow \mathbf{Q}).$$

3. **Disjunction ($P \vee Q$):** By De Morgan's Law: $P \vee Q \iff \neg(\neg P \wedge \neg Q) \iff (\neg P) \uparrow (\neg Q)$. Substituting $\neg P = P \uparrow P$ and $\neg Q = Q \uparrow Q$:

$$\mathbf{P \vee Q} \iff (\mathbf{P \uparrow P}) \uparrow (\mathbf{Q \uparrow Q}).$$

Since $\{\neg, \wedge, \vee\}$ is functionally complete, any Boolean truth function can be synthesized using exclusively the Sheffer Stroke $\uparrow$. ■

Worked Example 1.3 · Non-Definability via Automorphisms: Ordering in the Group of Integers

Consider the structure of integers under addition $\mathcal{Z} = (\mathbb{Z}, +)$. Prove that the standard order relation $<$ is **not first-order definable without parameters** in $\mathcal{Z}$.

Solution and Proof. Assume for contradiction that there exists a first-order formula $\varphi(x, y)$ in the language $\mathcal{L} = \{+\}$ such that for all $a, b \in \mathbb{Z}$:

$$a < b \iff \mathcal{Z} \models \varphi[a, b].$$

1. **Construct an Automorphism:** Consider the negation map $\pi : \mathbb{Z} \to \mathbb{Z}$ defined by $\pi(x) = -x$. We verify that π is an automorphism of $(\mathbb{Z}, +)$:
   - π is a bijection: $\pi(\pi(x)) = -(-x) = x$.
   - π preserves addition: $\pi(x + y) = -(x + y) = (-x) + (-y) = \pi(x) + \pi(y)$.

   Thus $\pi \in \mathrm{Aut}(\mathcal{Z})$.
2. **Apply the Automorphism Invariance Theorem:** If $\varphi(x, y)$ defines $<$, then by the Invariance Theorem:
$$\mathcal{Z} \models \varphi[a, b] \iff \mathcal{Z} \models \varphi[\pi(a), \pi(b)].$$
3. **Derive a Contradiction:** Choose $a = 1$ and $b = 2$. Since $1 < 2$, we have $\mathcal{Z} \models \varphi[1, 2]$. Applying the automorphism π:
$$\mathcal{Z} \models \varphi[\pi(1), \pi(2)] \iff \mathcal{Z} \models \varphi[-1, -2].$$
   Since φ defines the order relation $<$, this implies $-1 < -2$, which is mathematically false!

This contradiction proves that the order relation $<$ cannot be defined in the group $(\mathbb{Z}, +)$. ■

Worked Example 1.4 · Irrationality of $\sqrt{2}$ by Contradiction

Prove by contradiction that $\sqrt{2}$ is not a rational number ($\sqrt{2} \notin \mathbb{Q}$).

Solution and Proof. Assume to the contrary that the negation is true: $\sqrt{2} \in \mathbb{Q}$. By definition of the rational numbers $\mathbb{Q}$, there exist two integers $a, b \in \mathbb{Z}$ with $b \neq 0$ such that:

$$\sqrt{2} = \frac{a}{b}.$$

Without loss of generality, we assume the fraction is in **irreducible form**, meaning that $\gcd(a, b) = 1$ (a and b share no common positive divisor other than 1).
Squaring both sides of the equation:

$$2 = \frac{a^2}{b^2} \implies a^2 = 2b^2.$$

This equation implies that a^2 is an even integer (a multiple of 2).

- *Lemma:* If a^2 is even, then a is even.
- *Proof of Lemma by Contrapositive:* Suppose a is odd. Then $a = 2k + 1$ for some $k \in \mathbb{Z}$. Squaring gives $a^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. Since the contrapositive holds, the lemma is established.

Because a is even, there exists an integer $k \in \mathbb{Z}$ such that $a = 2k$. Substituting $a = 2k$ back into our equation $a^2 = 2b^2$:

$$(2k)^2 = 2b^2 \implies 4k^2 = 2b^2 \implies b^2 = 2k^2.$$

By the exact same reasoning, b^2 is even, which implies that b must also be an even integer. We have thus deduced that **both a and b are even**, meaning that 2 divides both a and b ($2 \mid \gcd(a, b)$). This directly contradicts our initial hypothesis that $\gcd(a, b) = 1$. Therefore, the initial assumption that $\sqrt{2} \in \mathbb{Q}$ is false, concluding the proof that:

$$\boldsymbol{\sqrt{2} \notin \mathbb{Q}.} \quad \blacksquare$$

Worked Example 1.5 · Proof by Induction: Sum of the First n Cubes

Prove by mathematical induction that for every positive integer $n \geq 1$:

$$\sum_{k=1}^{n} k^3 = 1^3 + 2^3 + 3^3 + \cdots + n^3 = \left[\frac{n(n+1)}{2}\right]^2.$$

Solution and Proof. Let $P(n)$ be the proposition: $\sum_{k=1}^{n} k^3 = \left[\frac{n(n+1)}{2}\right]^2$.
1. Base Step ($n = 1$): For $n = 1$:

$$\text{Left-Hand Side (LHS)} = 1^3 = 1.$$

$$\text{Right-Hand Side (RHS)} = \left[\frac{1(1+1)}{2}\right]^2 = \left[\frac{2}{2}\right]^2 = 1^2 = 1.$$

Since LHS = RHS = 1, $P(1)$ is true.
2. Inductive Step: Let $m \geq 1$ be an arbitrary integer. Assume that $P(m)$ is true (Induction Hypothesis):

$$\sum_{k=1}^{m} k^3 = \left[\frac{m(m+1)}{2}\right]^2.$$

We must show that $P(m + 1)$ is true, namely:

$$\sum_{k=1}^{m+1} k^3 = \left[\frac{(m+1)(m+2)}{2}\right]^2.$$

Starting from the definition of the sum of $m + 1$ terms:

$$\sum_{k=1}^{m+1} k^3 = \left(\sum_{k=1}^{m} k^3\right) + (m+1)^3.$$

Substituting the Induction Hypothesis:

$$\sum_{k=1}^{m+1} k^3 = \left[\frac{m(m+1)}{2}\right]^2 + (m+1)^3 = \frac{m^2(m+1)^2}{4} + (m+1)^3.$$

Factoring out the common factor $(m+1)^2$:

$$\sum_{k=1}^{m+1} k^3 = (m+1)^2\left[\frac{m^2}{4}+(m+1)\right] = (m+1)^2\left[\frac{m^2+4m+4}{4}\right]$$
$$= (m+1)^2\left[\frac{(m+2)^2}{4}\right] = \left[\frac{(m+1)(m+2)}{2}\right]^2.$$

This is precisely the statement $P(m+1)$.

3. Conclusion: By the Principle of Mathematical Induction, the identity holds for all integers $n \geq 1$:

$$\sum_{\mathbf{k=1}}^{\mathbf{n}} \mathbf{k^3} = \left[\frac{\mathbf{n(n+1)}}{\mathbf{2}}\right]^{\mathbf{2}} = \left(\sum_{\mathbf{k=1}}^{\mathbf{n}} \mathbf{k}\right)^{\mathbf{2}}. \quad \blacksquare$$

Worked Example 1.6 · Rigorous Proof of De Morgan's Law for Sets

Let A and B be two subsets of a universal set E. Prove rigorously by double inclusion that:

$$\complement_E(A \cap B) = \complement_E A \cup \complement_E B.$$

Solution and Proof. To establish set equality, we must prove two inclusions:

1. **Inclusion 1: $\complement_E(A \cap B) \subseteq \complement_E A \cup \complement_E B$.** Let $x \in \complement_E(A \cap B)$ be an arbitrary element. By definition of set complement:

   $$x \in E \quad \text{and} \quad x \notin (A \cap B).$$

   The statement $x \notin (A \cap B)$ is the logical negation of $x \in (A \cap B)$:

   $$\neg(x \in A \wedge x \in B).$$

   Applying De Morgan's logical law for conjunction:

   $$\neg(x \in A) \vee \neg(x \in B) \iff (x \notin A) \vee (x \notin B).$$

   Since $x \in E$, this translates to:

   $$(x \in E \wedge x \notin A) \vee (x \in E \wedge x \notin B) \iff x \in \complement_E A \vee x \in \complement_E B.$$

   By definition of set union:

   $$x \in (\complement_E A \cup \complement_E B).$$

   Since this holds for every $x \in \complement_E(A \cap B)$, we have established:

   $$\complement_E(A \cap B) \subseteq \complement_E A \cup \complement_E B.$$

2. **Inclusion 2: $\complement_E A \cup \complement_E B \subseteq \complement_E(A \cap B)$.** Let $y \in (\complement_E A \cup \complement_E B)$ be an arbitrary element. By definition of union:

   $$y \in \complement_E A \vee y \in \complement_E B \iff (y \in E \wedge y \notin A) \vee (y \in E \wedge y \notin B).$$

   Thus $y \in E$ and $(y \notin A \vee y \notin B)$. By De Morgan's logical law, $(y \notin A \vee y \notin B) \iff \neg(y \in A \wedge y \in B) \iff y \notin (A \cap B)$. Therefore:

   $$y \in E \quad \text{and} \quad y \notin (A \cap B) \implies y \in \complement_E(A \cap B).$$

   This establishes the reverse inclusion:

   $$\complement_E A \cup \complement_E B \subseteq \complement_E(A \cap B).$$

Combining both inclusions proves the identity:

$$\mathbf{\complement_E(A \cap B) = \complement_E A \cup \complement_E B}. \quad \blacksquare$$

Worked Example 1.7 · Characteristic Functions and Set Algebra

For any subset $A \subseteq E$, its **characteristic function** (or indicator) is defined as:

$$\mathbf{1}_A : E \longrightarrow \{0,1\}, \qquad \mathbf{1}_A(x) = \begin{cases} 1 & \text{if } x \in A, \\ 0 & \text{if } x \notin A. \end{cases}$$

1. Prove that for any two subsets $A, B \subseteq E$:

$$\mathbf{1}_{A\cap B} = \mathbf{1}_A \cdot \mathbf{1}_B, \qquad \mathbf{1}_{\complement_E A} = 1 - \mathbf{1}_A.$$

2. Using characteristic functions, derive the formula for the symmetric difference $\mathbf{1}_{A\Delta B}$.

Solution and Proof. 1. Intersection and Complement:

- For intersection: An element $x \in A \cap B \iff (x \in A \wedge x \in B)$. Thus $\mathbf{1}_{A\cap B}(x) = 1 \iff \mathbf{1}_A(x) = 1 \wedge \mathbf{1}_B(x) = 1 \iff \mathbf{1}_A(x) \cdot \mathbf{1}_B(x) = 1 \times 1 = 1$. If $x \notin A \cap B$, at least one of $\mathbf{1}_A(x)$ or $\mathbf{1}_B(x)$ is 0, making the product 0. Thus $\mathbf{1}_{A\cap B} = \mathbf{1}_A \cdot \mathbf{1}_B$.
- For complement: If $x \in \complement_E A$, then $x \notin A \implies \mathbf{1}_A(x) = 0$, so $1 - \mathbf{1}_A(x) = 1 - 0 = 1 = \mathbf{1}_{\complement_E A}(x)$. If $x \in A$, then $\mathbf{1}_A(x) = 1$, so $1 - \mathbf{1}_A(x) = 1 - 1 = 0 = \mathbf{1}_{\complement_E A}(x)$. Thus $\mathbf{1}_{\complement_E A} = 1 - \mathbf{1}_A$.

2. Symmetric Difference $\mathbf{1}_{A\Delta B}$: Recall that $A\Delta B = (A\setminus B)\cup(B\setminus A) = (A\cap\complement_E B)\cup(B\cap\complement_E A)$. Notice that $A \cap \complement_E B$ and $B \cap \complement_E A$ are strictly disjoint sets. For any two disjoint sets X and Y ($X \cap Y = \emptyset$), $\mathbf{1}_{X\cup Y} = \mathbf{1}_X + \mathbf{1}_Y$. Therefore:

$$\begin{aligned} \mathbf{1}_{A\Delta B} &= \mathbf{1}_{A\cap\complement_E B} + \mathbf{1}_{B\cap\complement_E A} \\ &= \mathbf{1}_A \cdot \mathbf{1}_{\complement_E B} + \mathbf{1}_B \cdot \mathbf{1}_{\complement_E A} \\ &= \mathbf{1}_A(1 - \mathbf{1}_B) + \mathbf{1}_B(1 - \mathbf{1}_A) \\ &= \mathbf{1}_A - \mathbf{1}_A\mathbf{1}_B + \mathbf{1}_B - \mathbf{1}_A\mathbf{1}_B \\ &= \mathbf{1_A} + \mathbf{1_B} - \mathbf{2 1_A 1_B} = (\mathbf{1}_A - \mathbf{1}_B)^2. \end{aligned}$$

This remarkable algebraic correspondence transforms boolean set manipulations directly into standard polynomial arithmetic!

Chapter 2

Binary Relations, Mappings & Abstract Algebraic Structures

2.1 Binary Relations and Equivalence Structures

Definition · Binary Relation and Fundamental Axiomatic Properties

Let E be a set. A **binary relation** $\mathcal{R}$ on E is formally defined as a subset of the Cartesian product: $\mathcal{R} \subseteq E \times E$. We write $x\mathcal{R}y$ if and only if $(x, y) \in \mathcal{R}$.

1. **Reflexivity:** $\forall x \in E,\ x\mathcal{R}x$.
2. **Symmetry:** $\forall x, y \in E,\ (x\mathcal{R}y \implies y\mathcal{R}x)$.
3. **Antisymmetry:** $\forall x, y \in E,\ ((x\mathcal{R}y \wedge y\mathcal{R}x) \implies x = y)$.
4. **Transitivity:** $\forall x, y, z \in E,\ ((x\mathcal{R}y \wedge y\mathcal{R}z) \implies x\mathcal{R}z)$.

Definition · Equivalence Relations and Quotient Sets

A binary relation $\mathcal{R}$ is an **equivalence relation** if it is simultaneously **reflexive, symmetric, and transitive**.

1. The **equivalence class** of an element $x \in E$ is the set of all elements related to x:
$$\mathrm{cl}(x) \equiv [x] = \{y \in E \mid x\mathcal{R}y\}.$$
2. The **quotient set** $E/\mathcal{R}$ is the set of all distinct equivalence classes:
$$E/\mathcal{R} = \{[x] \mid x \in E\}.$$

Theorem · Fundamental Theorem of Equivalence Relations (Partition Theorem)

Let $\mathcal{R}$ be an equivalence relation on E.

1. Two equivalence classes are either identical or completely disjoint:
$$\forall x, y \in E, \quad [x] = [y] \iff x\mathcal{R}y, \qquad [x] \cap [y] = \emptyset \iff \neg(x\mathcal{R}y).$$
2. The quotient set $E/\mathcal{R}$ forms a **partition** of E: the classes are non-empty, pairwise disjoint, and their union recovers E:
$$\bigcup_{[x] \in E/\mathcal{R}} [x] = E.$$

2.2 Order Relations and Posets

Definition · Partial and Total Orders

A binary relation $\leq$ on E is an **order relation** if it is **reflexive, antisymmetric, and transitive**. The pair $(E, \leq)$ is called a **partially ordered set** (or *poset*).

- **Total Order (Chain):** If every pair of elements is comparable:

$$\forall x, y \in E, \quad (x \leq y \vee y \leq x).$$

- **Partial Order:** There exist elements $x, y \in E$ such that neither $x \leq y$ nor $y \leq x$ (e.g. set inclusion $\subseteq$ on $\mathcal{P}(E)$).

2.3 Mappings, Functions, and Morphisms

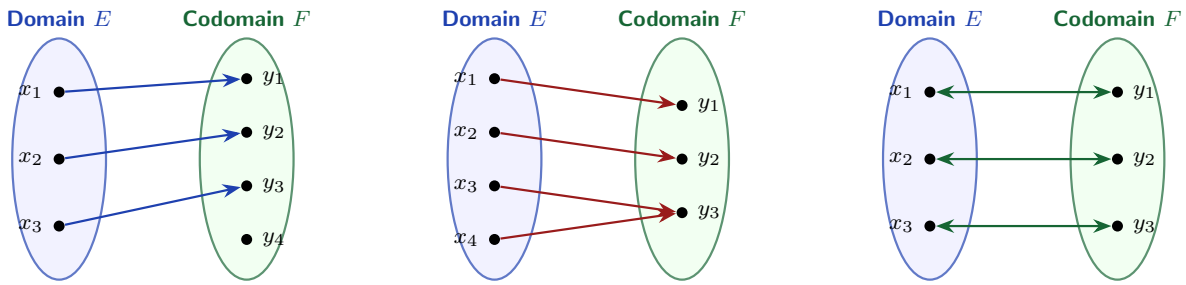


Figure 2.1: Classification of Mappings: Injective (each codomain element has at most one preimage), Surjective (every codomain element has at least one preimage), and Bijective (exact one-to-one correspondence).

Definition · Injectivity, Surjectivity, and Bijectivity

Let $f : E \longrightarrow F$ be a mapping from set E to set F:

1. **Injective (One-to-One):** Distinct domain elements map to distinct images:

$$\forall x_1, x_2 \in E, \quad (f(x_1) = f(x_2) \implies x_1 = x_2) \iff (x_1 \neq x_2 \implies f(x_1) \neq f(x_2)).$$

2. **Surjective (Onto):** Every element of the codomain is the image of at least one domain element:

$$\forall y \in F, \quad \exists x \in E \text{ such that } f(x) = y \iff f(E) = F.$$

3. **Bijective:** f is both injective and surjective. Equivalently, for every $y \in F$, the equation $f(x) = y$ has a **unique solution** $x \in E$:

$$\forall y \in F, \quad \exists! x \in E, \quad f(x) = y.$$

A mapping f is bijective if and only if there exists an inverse mapping $f^{-1} : F \longrightarrow E$ satisfying:

$$f^{-1} \circ f = \mathrm{id}_E \quad \text{and} \quad f \circ f^{-1} = \mathrm{id}_F.$$

Proposition / Lemma · Properties of Image and Preimage Sets

Let $f : E \longrightarrow F$. For any subsets $A, A' \subseteq E$ and $B, B' \subseteq F$:

1. $f(A \cup A') = f(A) \cup f(A')$ and $f(A \cap A') \subseteq f(A) \cap f(A')$ (equality holds if f is injective).
2. $f^{-1}(B \cup B') = f^{-1}(B) \cup f^{-1}(B')$ and $f^{-1}(B \cap B') = f^{-1}(B) \cap f^{-1}(B')$.
3. $f^{-1}(\complement_F B) = \complement_E(f^{-1}(B))$. (Preimages preserve all boolean operations!).

2.4 Abstract Algebraic Structures



Figure 2.2: Évariste Galois (1811–1832), French mathematical genius who introduced the concept of a *group* and symmetry permutations to resolve the unsolvability of general quintic polynomial equations by radicals.

Definition · Groups, Subgroups, and Homomorphisms

An **internal binary operation** on a set G is a mapping $* : G \times G \longrightarrow G, (x, y) \mapsto x * y$.

1. A pair $(G, *)$ is a **Group** if:
   - **Associativity:** $\forall x, y, z \in G,\ (x * y) * z = x * (y * z)$.
   - **Neutral Element:** $\exists e \in G,\ \forall x \in G,\ x * e = e * x = x$.
   - **Invertibility:** $\forall x \in G,\ \exists x' \in G,\ x * x' = x' * x = e$ (denoted x^{-1}).

   If additionally $\forall x, y \in G, x * y = y * x$, the group is **Abelian** (or commutative).
2. **Subgroup Criterion:** A non-empty subset $H \subseteq G$ is a subgroup ($H \leq G$) if and only if:
$$\forall x, y \in H, \quad x * y^{-1} \in H.$$
3. **Group Homomorphism:** A mapping $\phi : (G, *) \longrightarrow (G', \star)$ such that:
$$\forall x, y \in G, \quad \phi(x * y) = \phi(x) \star \phi(y).$$
   The **kernel** $\ker(\phi) = \{x \in G \mid \phi(x) = e'\}$ is a normal subgroup of G, and ϕ is injective if and only if $\ker(\phi) = \{e\}$.

Definition · Rings and Fields

1. A triplet $(A, +, \cdot)$ is a **Ring** if $(A, +)$ is an abelian group (neutral 0), the operation $\cdot$ is associative, and $\cdot$ distributes over $+$:

$$\forall x, y, z \in A, \quad x \cdot (y + z) = x \cdot y + x \cdot z \quad \text{and} \quad (y + z) \cdot x = y \cdot x + z \cdot x.$$

2. A ring $(K, +, \cdot)$ is a **Field** (Corps) if $0 \neq 1$, multiplication is commutative, and every non-zero element $x \in K \setminus \{0\}$ has a multiplicative inverse x^{-1}. (*Examples:* $\mathbb{Q}, \mathbb{R}, \mathbb{C}$ are fields; $\mathbb{Z}$ is an integral domain but not a field).

2.5 Worked Examples with Step-by-Step Proofs

Worked Example 2.1 · Modular Arithmetic as an Equivalence Relation

Fix an integer $n \geq 2$. On the set of integers $\mathbb{Z}$, define the relation of **congruence modulo** n by:

$$x \equiv y \pmod{n} \iff n \mid (x - y) \iff \exists k \in \mathbb{Z},\ x - y = kn.$$

1. Prove rigorously that $\equiv \pmod{n}$ is an equivalence relation on $\mathbb{Z}$.
2. Describe explicitly the equivalence classes and the quotient set $\mathbb{Z}/n\mathbb{Z}$.

Solution and Proof. 1. Verification of Equivalence Axioms:

- **Reflexivity:** For any $x \in \mathbb{Z}$, $x - x = 0 = 0 \cdot n$. Since $0 \in \mathbb{Z}$, $n \mid (x - x)$, so $x \equiv x \pmod{n}$.
- **Symmetry:** Let $x, y \in \mathbb{Z}$ such that $x \equiv y \pmod{n}$. Then $x - y = kn$ for some $k \in \mathbb{Z}$. Multiplying by -1:
$$y - x = -(x - y) = -kn = (-k)n.$$
Since $-k \in \mathbb{Z}$, $n \mid (y - x)$, so $y \equiv x \pmod{n}$.
- **Transitivity:** Let $x, y, z \in \mathbb{Z}$ such that $x \equiv y \pmod{n}$ and $y \equiv z \pmod{n}$. Then $x - y = k_1 n$ and $y - z = k_2 n$ for integers $k_1, k_2 \in \mathbb{Z}$. Adding the two equations:
$$(x - y) + (y - z) = k_1 n + k_2 n \implies x - z = (k_1 + k_2)n.$$
Since $k_1 + k_2 \in \mathbb{Z}$, $n \mid (x - z)$, so $x \equiv z \pmod{n}$.

Therefore, $\equiv \pmod{n}$ is an equivalence relation.
2. Equivalence Classes and Quotient Set: By the Euclidean division algorithm, for any integer $x \in \mathbb{Z}$, dividing by n yields unique integers $q \in \mathbb{Z}$ and $r \in \{0, 1, 2, \ldots, n - 1\}$ such that:

$$x = qn + r \implies x - r = qn \implies x \equiv r \pmod{n}.$$

Hence, every integer belongs to the equivalence class of its remainder $r \in \{0, 1, \ldots, n - 1\}$:

$$[r] = \{r + kn \mid k \in \mathbb{Z}\} = r + n\mathbb{Z}.$$

The quotient set consists of exactly n distinct equivalence classes:

$$\mathbb{Z}/\mathbf{n}\mathbb{Z} = \{[\mathbf{0}], [\mathbf{1}], [\mathbf{2}], \ldots, [\mathbf{n - 1}]\}. \quad \blacksquare$$

Worked Example 2.2 · Injectivity, Surjectivity, and Bijectivity Verification

Consider the mapping:

$$f : \mathbb{R} \setminus \{2\} \longrightarrow \mathbb{R} \setminus \{1\}, \qquad f(x) = \frac{x+1}{x-2}.$$

1. Prove from first principles that f is injective.
2. Prove from first principles that f is surjective.
3. Deduce that f is bijective and determine its explicit inverse mapping $f^{-1}(y)$.

Solution and Proof. **1. Proof of Injectivity:** Let $x_1, x_2 \in \mathbb{R} \setminus \{2\}$ such that $f(x_1) = f(x_2)$:

$$\frac{x_1+1}{x_1-2} = \frac{x_2+1}{x_2-2}.$$

Cross-multiplying denominators:

$$(x_1+1)(x_2-2) = (x_2+1)(x_1-2).$$

Expanding both products:

$$x_1x_2 - 2x_1 + x_2 - 2 = x_1x_2 - 2x_2 + x_1 - 2.$$

Subtracting the common terms $x_1x_2 - 2$ from both sides:

$$-2x_1 + x_2 = -2x_2 + x_1 \implies 3x_2 = 3x_1 \implies x_1 = x_2.$$

Thus f is strictly **injective**.
2. Proof of Surjectivity: Let $y \in \mathbb{R} \setminus \{1\}$ be an arbitrary element of the codomain. We must find $x \in \mathbb{R} \setminus \{2\}$ such that $f(x) = y$:

$$\frac{x+1}{x-2} = y \implies x + 1 = y(x-2) = yx - 2y.$$

Isolating terms in x:

$$1 + 2y = yx - x = x(y-1).$$

Since $y \neq 1$, we can divide by $(y-1) \neq 0$:

$$x = \frac{2y+1}{y-1}.$$

We must verify that this candidate $x \neq 2$: If $\frac{2y+1}{y-1} = 2 \implies 2y+1 = 2y-2 \implies 1 = -2$, which is impossible. Hence $x \in \mathbb{R} \setminus \{2\}$ always exists and is well-defined. Thus f is **surjective**.
3. Bijectivity and Inverse Mapping: Since f is both injective and surjective, it is a **bijection**. Its inverse mapping $f^{-1} : \mathbb{R} \setminus \{1\} \longrightarrow \mathbb{R} \setminus \{2\}$ is:

$$\mathbf{f^{-1}(y) = \frac{2y+1}{y-1}}. \quad \blacksquare$$

Worked Example 2.3 · Subgroup Verification and Matrix Groups

Let $\mathrm{GL}_2(\mathbb{R})$ denote the general linear group of 2×2 invertible real matrices under matrix multiplication. Consider the subset:

$$\mathrm{SL}_2(\mathbb{R}) = \{M \in \mathrm{GL}_2(\mathbb{R}) \mid \det(M) = 1\}.$$

Prove using the subgroup criterion that $\mathrm{SL}_2(\mathbb{R})$ is a subgroup of $\mathrm{GL}_2(\mathbb{R})$ (the *Special Linear Group*).

Solution and Proof. We verify the three conditions of the subgroup criterion:

1. **Non-emptiness:** The identity matrix $I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$ has determinant:

   $$\det(I_2) = (1)(1) - (0)(0) = 1.$$

   Thus $I_2 \in \mathrm{SL}_2(\mathbb{R})$, establishing that $\mathrm{SL}_2(\mathbb{R}) \neq \emptyset$.

2. **Closure under Inverse:** Let $A \in \mathrm{SL}_2(\mathbb{R})$. Then $\det(A) = 1 \neq 0$, so A^{-1} exists in $\mathrm{GL}_2(\mathbb{R})$. By the multiplicative property of determinants:

   $$\det(A \cdot A^{-1}) = \det(I_2) = 1 \implies \det(A) \cdot \det(A^{-1}) = 1.$$

   Since $\det(A) = 1$, we have $1 \cdot \det(A^{-1}) = 1 \implies \det(A^{-1}) = 1$. Therefore, $A^{-1} \in \mathrm{SL}_2(\mathbb{R})$.

3. **Closure under Product (Combined Criterion $AB^{-1} \in \mathrm{SL}_2(\mathbb{R})$):** Let $A, B \in \mathrm{SL}_2(\mathbb{R})$.

   $$\det(AB^{-1}) = \det(A) \cdot \det(B^{-1}) = 1 \times 1 = 1.$$

   Therefore, $AB^{-1} \in \mathrm{SL}_2(\mathbb{R})$.

By the Subgroup Criterion, $\mathrm{SL}_{\mathbf{2}}(\mathbb{R}) \leq \mathrm{GL}_{\mathbf{2}}(\mathbb{R})$ is a subgroup. ■

Worked Example 2.4 · Kernel and Image of a Group Homomorphism

Consider the mapping $\phi : (\mathbb{R}, +) \longrightarrow (\mathbb{C}^*, \cdot)$ defined by:

$$\phi(x) = e^{ix} = \cos(x) + i\sin(x).$$

1. Prove that ϕ is a group homomorphism.
2. Determine the kernel $\ker(\phi)$ and describe its algebraic nature.
3. Determine the image $\mathrm{Im}(\phi)$.

Solution and Proof. **1. Proof of Homomorphism:** For any $x, y \in \mathbb{R}$:

$$\phi(x + y) = e^{i(x+y)} = e^{ix} \cdot e^{iy} = \phi(x) \cdot \phi(y).$$

Since addition in the domain maps to multiplication in the codomain, ϕ is a group homomorphism.
2. Kernel Determination: The neutral element of $(\mathbb{C}^*, \cdot)$ is 1. By definition:

$$\ker(\phi) = \{x \in \mathbb{R} \mid \phi(x) = 1\} = \{x \in \mathbb{R} \mid \cos x + i\sin x = 1\}.$$

Equating real and imaginary parts:

$$\cos x = 1 \quad \text{and} \quad \sin x = 0 \iff x = 2k\pi \quad (k \in \mathbb{Z}).$$

Therefore:

$$\ker(\phi) = \mathbf{2}\pi\mathbb{Z} = \{\mathbf{2k}\pi \mid \mathbf{k} \in \mathbb{Z}\}.$$

Notice that $\ker(\phi) \neq \{0\}$, proving that ϕ is **not injective**.

3. Image Determination: For every $x \in \mathbb{R}$, $|\phi(x)| = |e^{ix}| = \sqrt{\cos^2 x + \sin^2 x} = 1$. Every complex number on the unit circle $\mathbb{U} = \{z \in \mathbb{C} \mid |z| = 1\}$ can be written in polar form as $e^{i\theta}$ for some $\theta \in \mathbb{R}$. Therefore:

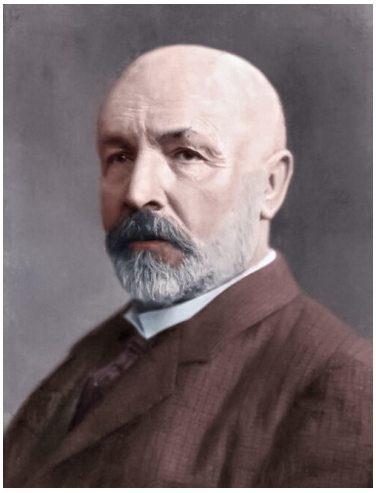
$$\mathrm{Im}(\phi) = \mathbb{U} \equiv \mathbb{S}^{\mathbf{1}} \quad \text{(the Circle Group)}. \quad \blacksquare$$

Chapter 3

The Real Number System ($\mathbb{R}$)

3.1 Axiomatic Structure of the Real Field

The set of real numbers $\mathbb{R}$ is axiomatically characterized as a **complete, Archimedean, totally ordered field**. While the rational numbers $\mathbb{Q}$ form an ordered field, they exhibit fatal topological "gaps" (such as $\sqrt{2} \notin \mathbb{Q}$ or $\pi \notin \mathbb{Q}$). The construction of $\mathbb{R}$ seals every gap through the **Completeness Axiom**.



(a) **Georg Cantor (1845–1918)**



(b) **Richard Dedekind (1831–1916)**

Figure 3.1: Founders of Real Analysis: Georg Cantor (inventor of transfinite set theory and diagonal cardinality) and Richard Dedekind (creator of Dedekind cuts, providing the first rigorous construction of the real continuum $\mathbb{R}$ from $\mathbb{Q}$).

Definition · Absolute Value and Distance on $\mathbb{R}$

For any real number $x \in \mathbb{R}$, its **absolute value** $|x|$ is defined by:

$$|x| = \begin{cases} x & \text{if } x \geq 0, \\ -x & \text{if } x < 0. \end{cases} \iff |x| = \max(x, -x) = \sqrt{x^2}.$$

The Euclidean distance between two numbers $x, y \in \mathbb{R}$ is $d(x, y) = |x - y|$.

Theorem · The Triangle Inequalities

For all real numbers $x, y \in \mathbb{R}$:

1. **Standard Triangle Inequality:**

$$|x + y| \leq |x| + |y|.$$

Equality holds if and only if x and y have the same sign ($xy \geq 0$).

2. **Reverse Triangle Inequality:**

$$||x| - |y|| \leq |x - y|.$$

3. **Generalized n-term Inequality:**

$$\left|\sum_{k=1}^{n} x_k\right| \leq \sum_{k=1}^{n} |x_k|.$$

3.2 Bounded Sets, Maximum, Minimum, Supremum and Infimum

Definition · Upper Bounds, Lower Bounds, Extrema

Let A be a non-empty subset of $\mathbb{R}$ ($A \subseteq \mathbb{R}, A \neq \emptyset$):

1. **Bounded Above (Majoré):** $\exists M \in \mathbb{R},\ \forall x \in A,\ x \leq M$. M is an **upper bound** (majorant).
2. **Bounded Below (Minoré):** $\exists m \in \mathbb{R},\ \forall x \in A,\ x \geq m$. m is a **lower bound** (minorant).
3. **Bounded Set (Borné):** A is both bounded above and below:

$$\exists K > 0, \quad \forall x \in A, \quad |x| \leq K.$$

4. **Maximum (Greatest Element):** An upper bound M that **belongs to the set**:

$$M = \max A \iff (M \in A \ \wedge\ \forall x \in A, x \leq M).$$

5. **Minimum (Least Element):** A lower bound m that belongs to the set:

$$m = \min A \iff (m \in A \ \wedge\ \forall x \in A, x \geq m).$$

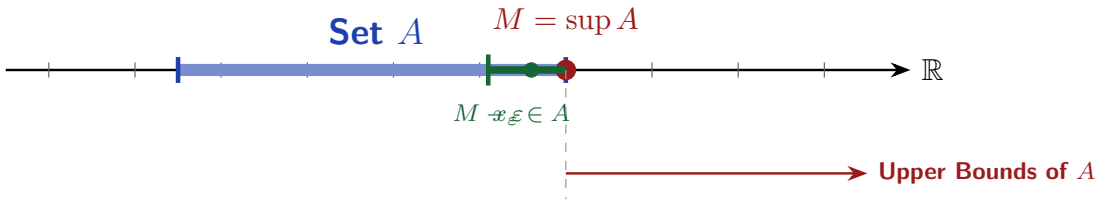


Figure 3.2: The Completeness Axiom and Supremum Characterization: The supremum $M = \sup A$ is the least upper bound. For any arbitrary small $\varepsilon > 0$, the interval $(M - \varepsilon, M]$ contains at least one element $x_\varepsilon \in A$.

Theorem · The Axiom of Completeness (Axiom of the Supremum)

Every non-empty subset $A \subset \mathbb{R}$ that is bounded above possesses a **least upper bound** in $\mathbb{R}$, denoted $\sup A$ (**supremum** or *borne supérieure*).
Similarly, every non-empty subset $A \subset \mathbb{R}$ that is bounded below possesses a **greatest lower bound** in $\mathbb{R}$, denoted $\inf A$ (**infimum** or *borne inférieure*).

Proposition / Lemma · Analytical ε-Characterization of Supremum and Infimum

Let $A \subset \mathbb{R}$ be non-empty and bounded above. A real number $S \in \mathbb{R}$ is the supremum ($S = \sup A$) if and only if:

$$\begin{cases} (1) & \forall x \in A, \quad x \leq S \quad (S \text{ is an upper bound}) \\ (2) & \forall \varepsilon > 0, \quad \exists x_\varepsilon \in A \text{ such that } x_\varepsilon > S - \varepsilon \quad (\text{no smaller upper bound exists}) \end{cases} \tag{3.1}$$

Similarly, $I = \inf A$ if and only if:

$$\begin{cases} (1) & \forall x \in A, \quad x \geq I \quad (I \text{ is a lower bound}) \\ (2) & \forall \varepsilon > 0, \quad \exists x_\varepsilon \in A \text{ such that } x_\varepsilon < I + \varepsilon \end{cases} \tag{3.2}$$

3.3 The Archimedean Property and Density

Theorem · The Archimedean Property of $\mathbb{R}$

The real number field $\mathbb{R}$ is **Archimedean**: it contains no infinitely large or infinitesimally small elements.

$$\forall x \in \mathbb{R}, \quad \exists n \in \mathbb{N} \quad \text{such that} \quad n > x. \tag{3.3}$$

Fundamental Analytical Equivalences:

1. $\forall \varepsilon > 0,\ \exists n \in \mathbb{N}^*$ such that $\frac{1}{n} < \varepsilon$.
2. $\forall x, y \in \mathbb{R}$ with $x > 0$, $\exists n \in \mathbb{N}$ such that $nx > y$.
3. The set of natural numbers $\mathbb{N}$ is not bounded above in $\mathbb{R}$ ($\sup \mathbb{N} = +\infty$).

Proposition / Lemma · The Floor Function (Integer Part)

For every real number $x \in \mathbb{R}$, there exists a **unique** integer $p \in \mathbb{Z}$ such that:

$$p \leq x < p + 1. \tag{3.4}$$

This integer is the **integer part** (or floor) of x, denoted $E(x) \equiv \lfloor x \rfloor = p$.

Theorem · Density of Rational and Irrational Numbers in $\mathbb{R}$

Between any two distinct real numbers, there exist infinitely many rational numbers and infinitely many irrational numbers:

$$\forall a, b \in \mathbb{R} \text{ with } a < b, \quad \exists r \in \mathbb{Q} \text{ and } \exists \alpha \in \mathbb{R} \setminus \mathbb{Q} \quad \text{such that} \quad a < r < \alpha < b. \tag{3.5}$$

In topological terms, both $\mathbb{Q}$ and $\mathbb{R} \setminus \mathbb{Q}$ are **dense** in $\mathbb{R}$ ($\overline{\mathbb{Q}} = \mathbb{R}$ and $\overline{\mathbb{R} \setminus \mathbb{Q}} = \mathbb{R}$).

3.4 Worked Examples with Step-by-Step Proofs

Worked Example 3.1 · Rigorous Proof of the Archimedean Property from Completeness

Prove rigorously that the Archimedean property ($\forall x \in \mathbb{R}, \exists n \in \mathbb{N} : n > x$) is a direct consequence of the Completeness Axiom of $\mathbb{R}$.

Solution and Proof. We proceed by contradiction (*reductio ad absurdum*). Assume that the Archimedean property is false. Then its logical negation is true:

$$\exists x_0 \in \mathbb{R}, \quad \forall n \in \mathbb{N}, \quad n \leq x_0.$$

This statement asserts that x_0 is an **upper bound** for the set of natural numbers $\mathbb{N} \subset \mathbb{R}$. Therefore, $\mathbb{N}$ is a non-empty subset of $\mathbb{R}$ bounded above by x_0.
By the **Completeness Axiom**, $\mathbb{N}$ must possess a supremum in $\mathbb{R}$:

$$S = \sup \mathbb{N} \in \mathbb{R}.$$

Now consider $\varepsilon = 1 > 0$. By the ε-characterization of the supremum, $S - 1$ is strictly strictly smaller than S, so $S - 1$ **cannot be an upper bound** of $\mathbb{N}$. Therefore, there exists at least one element $m \in \mathbb{N}$ such that:

$$m > S - 1.$$

Adding 1 to both sides of this inequality:

$$m + 1 > S.$$

Since $m \in \mathbb{N}$, its successor $m + 1$ is also a natural number ($m + 1 \in \mathbb{N}$). Thus, we have found an element of $\mathbb{N}$ that is strictly strictly greater than $S = \sup \mathbb{N}$. This contradicts the fundamental definition of S as an upper bound of $\mathbb{N}$!
Therefore, the assumption is impossible, establishing that:

$$\forall \mathbf{x} \in \mathbb{R}, \quad \exists \mathbf{n} \in \mathbb{N} \quad \text{such that} \quad \mathbf{n} > \mathbf{x}. \quad \blacksquare$$

Worked Example 3.2 · Determining $\sup, \inf, \max, \min$ of a Real Set

Consider the set:

$$A = \left\{ \frac{2n-1}{n+2} \;\middle|\; n \in \mathbb{N}^* \right\} = \left\{ \frac{1}{3}, \frac{3}{4}, \frac{5}{5}, \frac{7}{6}, \dots \right\}.$$

1. Determine $\min A$ and $\inf A$, proving your answer rigorously.
2. Prove that A has no maximum ($\max A$ does not exist).
3. Determine $\sup A$ using the ε-characterization property.

Solution and Proof. Rewrite the general element $x_n = \frac{2n-1}{n+2}$ by polynomial division:

$$x_n = \frac{2(n+2) - 5}{n+2} = 2 - \frac{5}{n+2}.$$

1. Infimum and Minimum: As n increases ($n \in \{1, 2, 3, \dots\}$), the denominator $n+2$ increases, so $\frac{5}{n+2}$ strictly decreases. Therefore, $x_n = 2 - \frac{5}{n+2}$ is a **strictly increasing sequence**:

$$x_1 = 2 - \frac{5}{3} = \frac{1}{3}, \quad x_2 = 2 - \frac{5}{4} = \frac{3}{4}, \quad \dots$$

For all $n \geq 1$:

$$x_n \geq x_1 = \frac{1}{3}.$$

Thus $1/3$ is a lower bound of A. Moreover, since $1/3 = x_1 \in A$:

$$\min \mathbf{A} = \inf \mathbf{A} = \boxed{\frac{1}{3}}.$$

2. Non-existence of Maximum: Since $n + 2 > 0$ for all $n \geq 1$, $\frac{5}{n+2} > 0$, which means:

$$x_n = 2 - \frac{5}{n+2} < 2 \quad \text{for all } n \in \mathbb{N}^*.$$

Thus 2 is an upper bound of A, but $2 \notin A$ (since $2 - \frac{5}{n+2} = 2 \implies \frac{5}{n+2} = 0$, which has no real solution). Because the sequence x_n is strictly increasing, every element x_n is strictly smaller than $x_{n+1} \in A$. Therefore, no single element in A can be the greatest element: $\max \mathbf{A}$ does not exist.

3. Supremum Determination via ε-Characterization: We claim that $\sup A = 2$.

- **Condition 1 (Upper Bound):** We already showed $\forall n \geq 1, x_n < 2$. Thus 2 is an upper bound.
- **Condition 2 (ε-Condition):** Let $\varepsilon > 0$ be given. We must find an integer $n \in \mathbb{N}^*$ such that:

$$x_n > 2 - \varepsilon \iff 2 - \frac{5}{n+2} > 2 - \varepsilon \iff \frac{5}{n+2} < \varepsilon \iff n + 2 > \frac{5}{\varepsilon} \iff n > \frac{5}{\varepsilon} - 2.$$

  By the Archimedean property of $\mathbb{R}$, there always exists an integer $n_0 \in \mathbb{N}^*$ such that $n_0 > \frac{5}{\varepsilon} - 2$. Choosing this n_0, we have $x_{n_0} \in A$ and $x_{n_0} > 2 - \varepsilon$.

Both conditions of the ε-characterization are satisfied. Therefore:

$$\sup \mathbf{A} = \mathbf{2}. \quad \blacksquare$$

Worked Example 3.3 · Density Proof: Constructing a Rational Between Any Two Reals

Prove rigorously that for any real numbers $x, y \in \mathbb{R}$ with $x < y$, there exists a rational number $r \in \mathbb{Q}$ such that:

$$x < r < y.$$

Solution and Proof. Since $x < y$, the interval length is strictly positive: $y - x > 0$.

1. **Choice of Denominator n:** By the Archimedean property, there exists an integer $n \in \mathbb{N}^*$ such that:
$$n > \frac{1}{y - x} \iff \frac{1}{n} < y - x \iff ny - nx > 1.$$
2. **Choice of Numerator m:** Consider the number nx. By the floor function property, there exists a unique integer $m \in \mathbb{Z}$ such that:
$$m - 1 \leq nx < m \iff nx < m \leq nx + 1.$$
3. **Verification of Inequalities:** From the left inequality:
$$nx < m \implies x < \frac{m}{n}.$$

From the right inequality and since $ny - nx > 1 \implies nx + 1 < ny$:

$$m \leq nx + 1 < ny \implies m < ny \implies \frac{m}{n} < y.$$

Combining both inequalities:

$$x < \frac{m}{n} < y.$$

Setting $r = \frac{m}{n} \in \mathbb{Q}$, we have established that $\mathbf{x < r < y}$. ■

Worked Example 3.4 · Supremum of a Two-Variable Bounded Set

Let A and B be two non-empty subsets of $\mathbb{R}$ bounded above. Define:

$$A + B = \{x + y \mid x \in A \wedge y \in B\}.$$

Prove that $A + B$ is bounded above and that:

$$\sup(A + B) = \sup A + \sup B.$$

Solution and Proof. Let $S_A = \sup A$ and $S_B = \sup B$.

1. **$S_A + S_B$ is an Upper Bound for $A + B$:** Let $z \in A + B$. By definition, $z = x + y$ with $x \in A$ and $y \in B$. Since S_A is an upper bound of A, $x \leq S_A$. Since S_B is an upper bound of B, $y \leq S_B$. Adding the two inequalities:

   $$z = x + y \leq S_A + S_B.$$

   Since this holds for every $z \in A+B$, $A+B$ is bounded above by S_A+S_B. By the Completeness Axiom, $\sup(A + B)$ exists, and:

   $$\sup(A + B) \leq S_A + S_B.$$

2. **Reverse Inequality via ε-Characterization:** Let $\varepsilon > 0$ be given. Consider $\varepsilon' = \frac{\varepsilon}{2} > 0$.

   - By the ε-characterization of $\sup A$, there exists $x_0 \in A$ such that $x_0 > S_A - \frac{\varepsilon}{2}$.
   - By the ε-characterization of $\sup B$, there exists $y_0 \in B$ such that $y_0 > S_B - \frac{\varepsilon}{2}$.

   Adding these two inequalities:

   $$x_0 + y_0 > \left(S_A - \frac{\varepsilon}{2}\right) + \left(S_B - \frac{\varepsilon}{2}\right) = (S_A + S_B) - \varepsilon.$$

   Since $x_0 + y_0 \in A + B$, we have found an element of $A + B$ strictly greater than $(S_A + S_B) - \varepsilon$. Therefore, no number strictly smaller than $S_A + S_B$ can be an upper bound for $A + B$. This proves that:

   $$\sup(A + B) \geq S_A + S_B.$$

Combining both inequalities yields the exact equality:

$$\sup(\mathbf{A} + \mathbf{B}) = \sup \mathbf{A} + \sup \mathbf{B}. \quad ■$$

Chapter 4

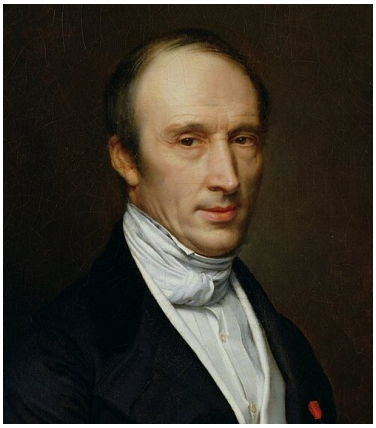
Numerical Sequences & Convergence

4.1 The Limit Concept and the ε-N Definition

A **numerical sequence** of real numbers is a mapping from the natural numbers into the reals:

$$u : \mathbb{N} \longrightarrow \mathbb{R}, \qquad n \longmapsto u(n) \equiv u_n.$$

We denote the entire sequence by $(u_n)_{n\in\mathbb{N}}$ or simply (u_n).



(a) **Augustin-Louis Cauchy (1789–1857)**



(b) **Bernard Bolzano (1781–1848)**

Figure 4.1: Architects of Modern Rigorous Analysis: Augustin-Louis Cauchy (who formalized ε-δ limits, continuity, and Cauchy sequences) and Bernard Bolzano (pioneer of the Bolzano-Weierstrass theorem and intermediate value theorem).

Definition · Convergence of a Sequence (ε-N Formulation)

A sequence (u_n) is said to **converge** to a finite real limit $l \in \mathbb{R}$, denoted $\lim_{n\to\infty} u_n = l$ or $u_n \xrightarrow[n\to\infty]{} l$, if:

$$\forall \varepsilon > 0, \quad \exists N_\varepsilon \in \mathbb{N} \quad \text{such that} \quad \forall n \in \mathbb{N}, \quad (n > N_\varepsilon \implies |u_n - l| < \varepsilon)\,. \tag{4.1}$$

Equivalently: for every interval $(l-\varepsilon, l+\varepsilon)$ centered on l, all terms of the sequence from index $N_\varepsilon + 1$ onward lie strictly inside the interval. A sequence that does not converge is called **divergent**.

Theorem · Uniqueness of the Limit and Boundedness

1. **Uniqueness:** If a sequence (u_n) converges, its limit l is **unique**:

$$(u_n \to l_1 \quad \text{and} \quad u_n \to l_2) \implies l_1 = l_2.$$

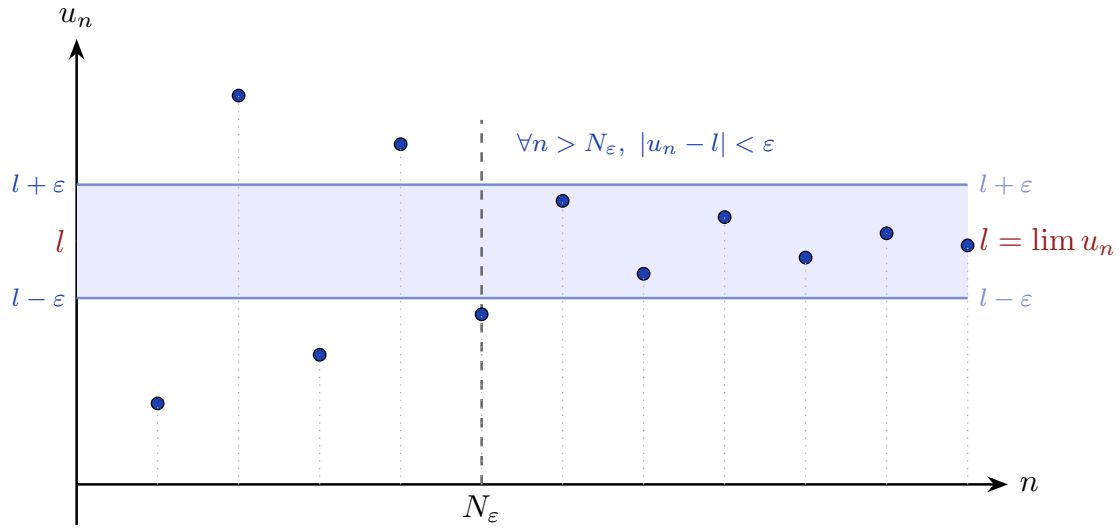


Figure 4.2: Geometric Interpretation of Sequence Convergence: Given an arbitrary tolerance $\varepsilon > 0$, an index N_ε exists such that for all indices $n > N_\varepsilon$, all sequence points u_n are permanently trapped within the horizontal ε-tube $(l - \varepsilon, l + \varepsilon)$.

2. **Boundedness:** Every convergent sequence is **bounded**:

$$(u_n \to l) \implies \exists M > 0, \quad \forall n \in \mathbb{N}, \quad |u_n| \leq M.$$

(*Note:* The converse is false; e.g. $u_n = (-1)^n$ is bounded but divergent).

Definition · Infinite Limits and Divergence to Infinity

1. $\lim_{n\to\infty} u_n = +\infty \iff \forall A > 0,\ \exists N_A \in \mathbb{N},\ \forall n > N_A,\ u_n > A.$
2. $\lim_{n\to\infty} u_n = -\infty \iff \forall B > 0,\ \exists N_B \in \mathbb{N},\ \forall n > N_B,\ u_n < -B.$

4.2 Fundamental Convergence Criteria

Theorem · The Squeeze (Sandwich / Gendarmes) Theorem

Let $(u_n), (v_n), (w_n)$ be three real sequences such that:

1. There exists an integer $n_0 \in \mathbb{N}$ such that $\forall n \geq n_0,\ v_n \leq u_n \leq w_n$;
2. $\lim_{n\to\infty} v_n = \lim_{n\to\infty} w_n = l$.

Then the sequence (u_n) converges and its limit is:

$$\lim_{n\to\infty} u_n = l. \tag{4.2}$$

Theorem · The Monotone Convergence Theorem

1. If (u_n) is **monotonically increasing** $(u_{n+1} \geq u_n)$ and **bounded above**, then (u_n) converges, and its limit is the supremum of its range:

$$\lim_{n\to\infty} u_n = \sup\{u_n \mid n \in \mathbb{N}\}.$$

If it is increasing and not bounded above, then $\lim_{n\to\infty} u_n = +\infty$.

2. If (u_n) is **monotonically decreasing** ($u_{n+1} \leq u_n$) and **bounded below**, then (u_n) converges, and its limit is the infimum of its range:

$$\lim_{n\to\infty} u_n = \inf\{u_n \mid n \in \mathbb{N}\}.$$

If it is decreasing and not bounded below, then $\lim_{n\to\infty} u_n = -\infty$.

4.3 Subsequences, Cauchy Sequences, and Adjacent Sequences

Definition · Subsequences (Suites Extraites)

Let $(u_n)_{n\in\mathbb{N}}$ be a sequence. A **subsequence** is a sequence of the form $(u_{\phi(k)})_{k\in\mathbb{N}}$, where:

$$\phi : \mathbb{N} \longrightarrow \mathbb{N} \quad \text{is a strictly increasing function} \quad (\phi(0) < \phi(1) < \phi(2) < \dots).$$

Note that for every $k \in \mathbb{N}$, $\phi(k) \geq k$. If $u_n \to l$, then every subsequence converges to the exact same limit: $u_{\phi(k)} \xrightarrow[k\to\infty]{} l$.

Theorem · The Bolzano-Weierstrass Theorem for Sequences

Every **bounded sequence** of real numbers possesses at least one **convergent subsequence**.

Definition · Cauchy Sequences and Completeness

A sequence (u_n) is a **Cauchy sequence** if its terms become arbitrarily close to each other as their indices grow:

$$\forall \varepsilon > 0, \quad \exists N \in \mathbb{N} \quad \text{such that} \quad \forall p, q \in \mathbb{N}, \quad ((p > N \ \wedge \ q > N) \implies |u_p - u_q| < \varepsilon). \tag{4.3}$$

The Cauchy Criterion (Completeness of $\mathbb{R}$): In the real number field $\mathbb{R}$, a sequence converges to a finite limit if and only if it is a Cauchy sequence:

$$(u_n) \text{ converges in } \mathbb{R} \iff (u_n) \text{ is a Cauchy sequence.}$$

Definition · Adjacent Sequences (Suites Adjacentes)

Two sequences (u_n) and (v_n) are said to be **adjacent** if:

1. (u_n) is monotonically increasing;
2. (v_n) is monotonically decreasing;
3. $\lim_{n\to\infty}(v_n - u_n) = 0$.

Theorem on Adjacent Sequences: If (u_n) and (v_n) are adjacent, then:

- For all $n \in \mathbb{N}$, $u_n \leq v_n$;
- Both sequences converge to the **exact same common limit** $l \in \mathbb{R}$:

$$\lim_{n\to\infty} u_n = \lim_{n\to\infty} v_n = l, \qquad \text{and} \quad \forall n \in \mathbb{N}, \quad u_n \leq l \leq v_n.$$

4.4 Worked Examples with Step-by-Step Proofs

Worked Example 4.1 · Rigorous Proof of Limit Uniqueness by Contradiction

Prove from the formal ε-N definition that the limit of a convergent sequence is unique.

Solution and Proof. Let (u_n) be a sequence. Assume to the contrary that (u_n) converges to two distinct limits $l_1 \neq l_2$. Then the distance between them is strictly positive:

$$|l_1 - l_2| > 0.$$

Choose $\varepsilon = \frac{|l_1 - l_2|}{2} > 0$.

1. Since $u_n \to l_1$, by the definition of limit for this ε, there exists an integer $N_1 \in \mathbb{N}$ such that:

$$\forall n > N_1, \quad |u_n - l_1| < \varepsilon = \frac{|l_1 - l_2|}{2}.$$

2. Since $u_n \to l_2$, there exists an integer $N_2 \in \mathbb{N}$ such that:

$$\forall n > N_2, \quad |u_n - l_2| < \varepsilon = \frac{|l_1 - l_2|}{2}.$$

Now choose any index $n_0 > \max(N_1, N_2)$. For this n_0, both inequalities hold simultaneously. Using the triangle inequality:

$$\begin{aligned} |l_1 - l_2| &= |(l_1 - u_{n_0}) + (u_{n_0} - l_2)| \\ &\leq |l_1 - u_{n_0}| + |u_{n_0} - l_2| = |u_{n_0} - l_1| + |u_{n_0} - l_2| \\ &< \frac{|l_1 - l_2|}{2} + \frac{|l_1 - l_2|}{2} = |l_1 - l_2|. \end{aligned}$$

This yields the strict inequality:

$$|l_1 - l_2| < |l_1 - l_2|,$$

which is an undeniable logical contradiction ($x < x$). Therefore, the assumption that $l_1 \neq l_2$ is false, establishing that:

$$\mathbf{l_1 = l_2}. \quad \blacksquare$$

Worked Example 4.2 · Verification of Limit via Formal ε-N Argument

Using the formal ε-N definition of limit, prove that:

$$\lim_{n\to\infty} \frac{3n+5}{2n-1} = \frac{3}{2}.$$

Solution and Proof. Let $\varepsilon > 0$ be given. We must find an integer $N_\varepsilon \in \mathbb{N}$ such that for all $n > N_\varepsilon$:

$$\left|\frac{3n+5}{2n-1} - \frac{3}{2}\right| < \varepsilon.$$

Compute the algebraic difference over a common denominator:

$$\frac{3n+5}{2n-1} - \frac{3}{2} = \frac{2(3n+5) - 3(2n-1)}{2(2n-1)} = \frac{(6n+10) - (6n-3)}{2(2n-1)} = \frac{13}{2(2n-1)}.$$

For $n \geq 1$, $2n - 1 > 0$, so the absolute value can be removed:

$$\left|\frac{3n+5}{2n-1} - \frac{3}{2}\right| = \frac{13}{4n-2}.$$

We require:

$$\frac{13}{4n-2} < \varepsilon \iff 4n-2 > \frac{13}{\varepsilon} \iff 4n > \frac{13}{\varepsilon} + 2 \iff n > \frac{13}{4\varepsilon} + \frac{1}{2}.$$

By the Archimedean property of $\mathbb{R}$, we can choose:

$$N_\varepsilon = \left\lfloor \frac{13}{4\varepsilon} + \frac{1}{2} \right\rfloor + 1 \in \mathbb{N}.$$

Then for any $n > N_\varepsilon$:

$$n > \frac{13}{4\varepsilon} + \frac{1}{2} \implies 4n-2 > \frac{13}{\varepsilon} \implies \left| \frac{3n+5}{2n-1} - \frac{3}{2} \right| = \frac{13}{4n-2} < \varepsilon.$$

This completes the formal proof that $\lim_{\mathbf{n\to\infty}} \frac{\mathbf{3n+5}}{\mathbf{2n-1}} = \frac{\mathbf{3}}{\mathbf{2}}$. ■

Worked Example 4.3 · Convergence of a Recurring Sequence via Monotone Convergence

Consider the sequence $(u_n)_{n\in\mathbb{N}}$ defined by recurrence:

$$u_0 = 1, \qquad u_{n+1} = \sqrt{2+u_n} \quad \text{for all } n \geq 0.$$

1. Prove by mathematical induction that for all $n \in \mathbb{N}$, $1 \leq u_n < 2$.
2. Prove by mathematical induction that (u_n) is strictly increasing.
3. Deduce that (u_n) converges and calculate its limit l.

Solution and Proof. 1. Boundedness by Induction ($1 \leq u_n < 2$):

- **Base step ($n = 0$):** $u_0 = 1$, which satisfies $1 \leq 1 < 2$. True.
- **Inductive step:** Assume $1 \leq u_k < 2$ for some $k \geq 0$. Adding 2: $3 \leq 2 + u_k < 4$. Since the square root function is strictly increasing on $\mathbb{R}^+$:

$$\sqrt{3} \leq \sqrt{2+u_k} < \sqrt{4} = 2.$$

  Since $1 < \sqrt{3} \approx 1.732$, we have $1 \leq u_{k+1} < 2$.

Thus, by induction, $1 \leq u_n < 2$ for all $n \in \mathbb{N}$.
2. Monotonicity by Induction ($u_{n+1} > u_n$):

- **Base step ($n = 0$):** $u_0 = 1$ and $u_1 = \sqrt{2+1} = \sqrt{3} \approx 1.732 > 1 = u_0$. True.
- **Inductive step:** Assume $u_{k+1} > u_k$ for some $k \geq 0$. Adding 2: $2 + u_{k+1} > 2 + u_k$. Taking square roots:

$$\sqrt{2+u_{k+1}} > \sqrt{2+u_k} \implies u_{k+2} > u_{k+1}.$$

Thus, (u_n) is strictly increasing.
3. Convergence and Limit Calculation: Because (u_n) is strictly increasing and bounded above by 2, the **Monotone Convergence Theorem** guarantees that (u_n) converges to a finite limit $l \in [1, 2]$:

$$\lim_{n\to\infty} u_n = l.$$

Since the function $f(x) = \sqrt{2+x}$ is continuous on $[1, 2]$:

$$\lim_{n\to\infty} u_{n+1} = f\left(\lim_{n\to\infty} u_n\right) \implies l = \sqrt{2+l}.$$

Squaring both sides ($l \geq 1 > 0$):

$$l^2 = 2 + l \implies l^2 - l - 2 = 0 \implies (l-2)(l+1) = 0.$$

The roots are $l = 2$ and $l = -1$. Since $u_n \geq 1$ for all n, the limit must be positive ($l \geq 1$), which rejects $l = -1$. Therefore:

$$\lim_{\mathbf{n}\to\infty} \mathbf{u_n} = \mathbf{2}. \quad \blacksquare$$

Worked Example 4.4 · Euler's Constant e via Adjacent Sequences

Define the two sequences for $n \geq 1$:

$$u_n = \left(1 + \frac{1}{n}\right)^n, \qquad v_n = \left(1 + \frac{1}{n}\right)^{n+1}.$$

1. Show that $v_n = u_n\left(1 + \frac{1}{n}\right)$ and that $\lim_{n\to\infty}(v_n - u_n) = 0$.
2. Given that (u_n) is strictly increasing and (v_n) is strictly decreasing, prove that (u_n) and (v_n) are adjacent.
3. Conclude on the existence of Euler's number $e = \lim_{n\to\infty}\left(1 + \frac{1}{n}\right)^n$ and establish the framing $u_n < e < v_n$.

Solution and Proof. 1. Ratio and Difference:

$$v_n = \left(1 + \frac{1}{n}\right)^{n+1} = \left(1 + \frac{1}{n}\right)^n \cdot \left(1 + \frac{1}{n}\right) = u_n\left(1 + \frac{1}{n}\right).$$

The difference is:

$$v_n - u_n = u_n\left(1 + \frac{1}{n}\right) - u_n = \frac{u_n}{n}.$$

By the binomial theorem, $u_n = \sum_{k=0}^{n}\binom{n}{k}\frac{1}{n^k} = 1 + 1 + \frac{1}{2!}\left(1 - \frac{1}{n}\right) + \cdots < \sum_{k=0}^{\infty}\frac{1}{k!} = 3$. Since $0 < u_n < 3$:

$$0 < v_n - u_n = \frac{u_n}{n} < \frac{3}{n}.$$

Taking limits as $n \to \infty$, since $\lim \frac{3}{n} = 0$, by the squeeze theorem:

$$\lim_{n\to\infty}(v_n - u_n) = 0.$$

2. Adjacency of the Sequences:

- (u_n) is strictly increasing;
- (v_n) is strictly decreasing;
- $\lim_{n\to\infty}(v_n - u_n) = 0$.

All three conditions of the **Adjacent Sequences Theorem** are satisfied.
3. Existence and Definition of e: By the theorem on adjacent sequences, (u_n) and (v_n) converge to a unique common limit, which defines **Euler's number** e:

$$\lim_{\mathbf{n}\to\infty}\left(\mathbf{1} + \frac{\mathbf{1}}{\mathbf{n}}\right)^{\mathbf{n}} = \lim_{\mathbf{n}\to\infty}\left(\mathbf{1} + \frac{\mathbf{1}}{\mathbf{n}}\right)^{\mathbf{n+1}} = \mathbf{e} \approx \mathbf{2.718\,281\,828}\ldots$$

Furthermore, for every positive integer $n \geq 1$:

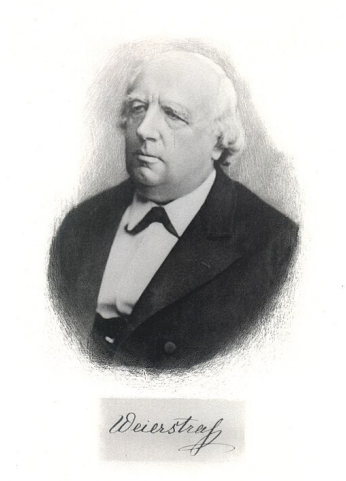
$$\left(\mathbf{1} + \frac{\mathbf{1}}{\mathbf{n}}\right)^{\mathbf{n}} < \mathbf{e} < \left(\mathbf{1} + \frac{\mathbf{1}}{\mathbf{n}}\right)^{\mathbf{n+1}}. \quad \blacksquare$$

Chapter 5

Real Functions of a Real Variable: Limits, Continuity & Derivatives

5.1 Limits of Real Functions and the ε-δ Definition

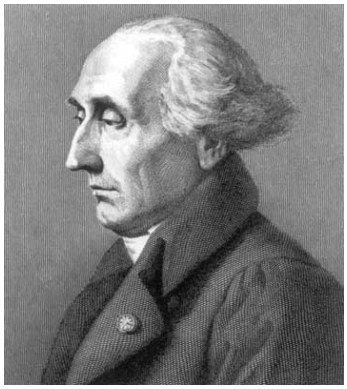
Let $I \subseteq \mathbb{R}$ be an interval and $f : I \longrightarrow \mathbb{R}$ a real function. Let $x_0 \in \mathbb{R}$ be an accumulation point of I.



(a) **Karl Weierstrass (1815–1897)**



(b) **Michel Rolle (1652–1719)**



(c) **Joseph-Louis Lagrange (1736–1813)**

Figure 5.1: Masters of Differential Calculus: Karl Weierstrass ("Father of Modern Analysis", creator of the static ε-δ definition), Michel Rolle (formulator of Rolle's Theorem in 1691), and Joseph-Louis Lagrange (discoverer of the Mean Value Theorem and remainder formulas for Taylor series).

Definition · Formal ε-δ Definition of Functional Limit

The function f has a **limit** $L \in \mathbb{R}$ as x approaches x_0, written $\lim_{x \to x_0} f(x) = L$, if:

$$\forall \varepsilon > 0, \quad \exists \delta > 0 \quad \text{such that} \quad \forall x \in I, \quad (0 < |x - x_0| < \delta \implies |f(x) - L| < \varepsilon)\,. \tag{5.1}$$

Theorem · Sequential Characterization of Functional Limits (Heine's Theorem)

Let $f : I \longrightarrow \mathbb{R}$ and x_0 an accumulation point of I. The following assertions are logically equivalent:

1. $\lim_{x \to x_0} f(x) = L$;
2. For every sequence $(x_n)_{n \in \mathbb{N}}$ in $I \setminus \{x_0\}$ converging to x_0, the image sequence $(f(x_n))$ converges to L:
$$\forall (x_n) \subset I \setminus \{x_0\}, \quad \Big(\lim_{n \to \infty} x_n = x_0 \implies \lim_{n \to \infty} f(x_n) = L\Big)\,.$$

Corollary (Disproof Criterion): If two sequences $x_n \to x_0$ and $x'_n \to x_0$ yield different limits $\lim f(x_n) \neq \lim f(x'_n)$, then $\lim_{x \to x_0} f(x)$ does not exist (e.g. $f(x) = \sin(1/x)$ as $x \to 0$).

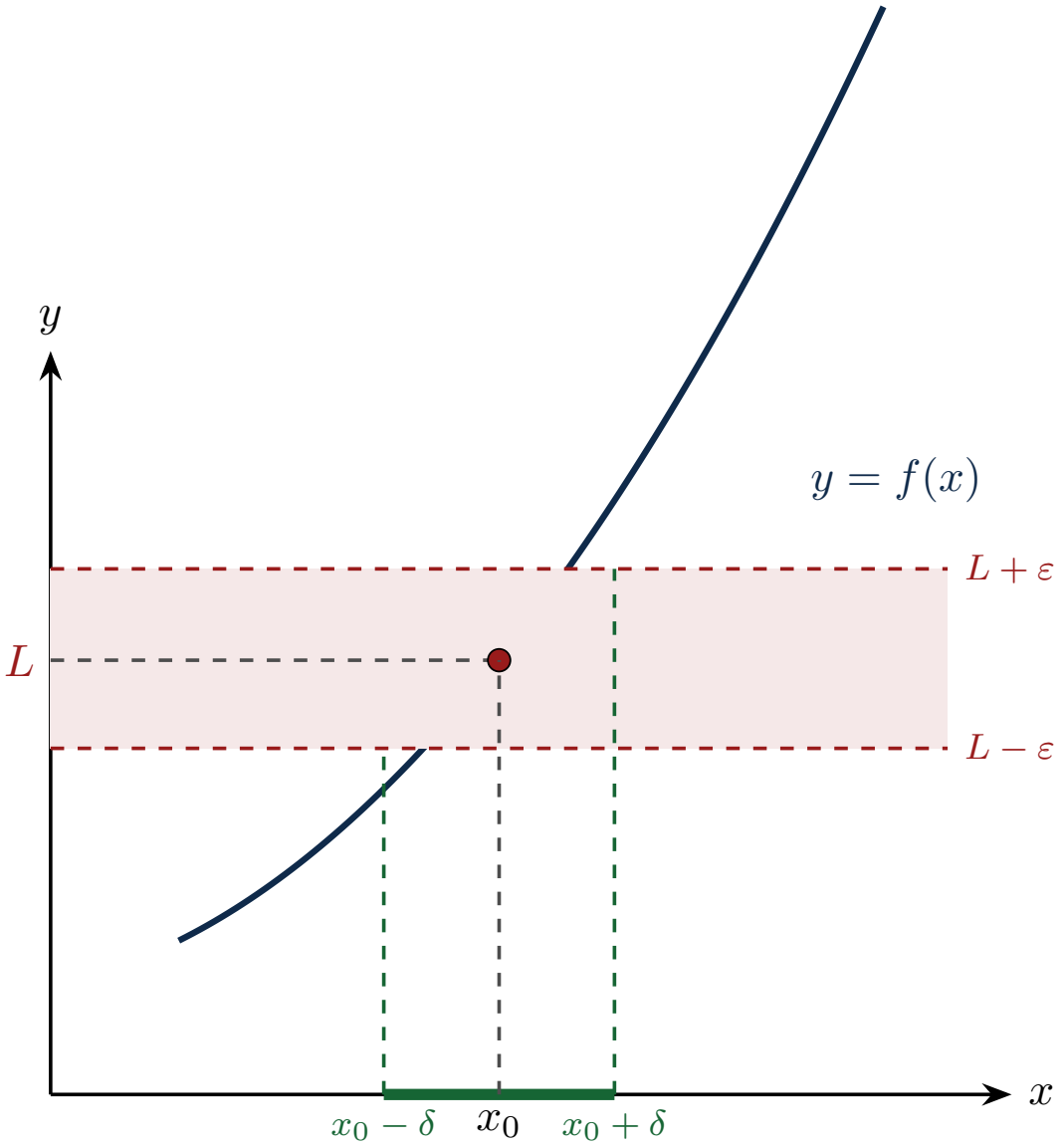


Figure 5.2: Weierstrass ε-δ Neighborhood Geometry: For any target band $(L - \varepsilon, L + \varepsilon)$ along the y-axis, there exists a surrounding interval $(x_0 - \delta, x_0 + \delta)$ on the x-axis whose functional images fall strictly within the band.

5.2 Continuity of Real Functions

Definition · Continuity at a Point and on an Interval

1. A function $f : I \longrightarrow \mathbb{R}$ is **continuous at a point** $x_0 \in I$ if:

$$\lim_{x \to x_0} f(x) = f(x_0) \iff \forall \varepsilon > 0,\ \exists \delta > 0,\ \forall x \in I,\ (|x - x_0| < \delta \implies |f(x) - f(x_0)| < \varepsilon).$$

2. f is **continuous on the interval** I if it is continuous at every point $x_0 \in I$. We write $f \in \mathcal{C}^0(I)$ or $f \in \mathcal{C}(I, \mathbb{R})$.

Theorem · The Intermediate Value Theorem (Théorème des Valeurs Intermédiaires)

Let $f : [a, b] \longrightarrow \mathbb{R}$ be a continuous function on a closed bounded interval $[a, b]$.

1. For every real number y lying between $f(a)$ and $f(b)$, there exists at least one real number $c \in [a, b]$ such that:

$$f(c) = y.$$

2. **Bolzano's Root Theorem:** If $f(a) \cdot f(b) < 0$ (opposite signs at endpoints), then there exists at least one root $c \in (a, b)$ such that $f(c) = 0$.

Theorem · The Extreme Value Theorem (Weierstrass's Theorem)

Let $f : [a, b] \longrightarrow \mathbb{R}$ be a continuous function on a compact (closed and bounded) interval $[a, b]$. Then f is **bounded** and **attains its bounds**:

$$\exists x_{\min}, x_{\max} \in [a, b] \quad \text{such that} \quad f(x_{\min}) = \inf_{x \in [a,b]} f(x) \quad \text{and} \quad f(x_{\max}) = \sup_{x \in [a,b]} f(x). \tag{5.2}$$

Thus, $f([a, b]) = [m, M]$ is itself a closed and bounded interval.

5.3 Differential Calculus: Definitions and Core Properties

Definition · The Derivative at a Point

Let $f : I \longrightarrow \mathbb{R}$ and x_0 an interior point of I. The function f is **differentiable at** x_0 if the difference quotient has a finite limit as $h \to 0$:

$$f'(x_0) = \lim_{h \to 0} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{x \to x_0} \frac{f(x) - f(x_0)}{x - x_0}. \tag{5.3}$$

$f'(x_0)$ is the **derivative** of f at x_0. Geometrically, $f'(x_0)$ is the exact slope of the tangent line to the curve $y = f(x)$ at $(x_0, f(x_0))$:

$$y = f(x_0) + f'(x_0)(x - x_0).$$

Theorem · Differentiability Implies Continuity

If f is differentiable at x_0, then f is necessarily continuous at x_0:

$$f \text{ differentiable at } x_0 \implies f \text{ continuous at } x_0. \tag{5.4}$$

(*Caution:* The converse is false! The absolute value function $f(x) = |x|$ is continuous at $x = 0$ but not differentiable at $x = 0$).

Proposition / Lemma · Algebra of Derivatives

Let $u, v : I \longrightarrow \mathbb{R}$ be differentiable at x:

1. **Linearity:** $(\alpha u + \beta v)'(x) = \alpha u'(x) + \beta v'(x)$ for all $\alpha, \beta \in \mathbb{R}$.
2. **Leibniz Product Rule:** $(u \cdot v)'(x) = u'(x)v(x) + u(x)v'(x)$.
3. **Quotient Rule:** $\left(\frac{u}{v}\right)'(x) = \frac{u'(x)v(x) - u(x)v'(x)}{[v(x)]^2}$ (provided $v(x) \neq 0$).
4. **Chain Rule:** If f is differentiable at x and g is differentiable at $f(x)$:
$$(g \circ f)'(x) = g'(f(x)) \cdot f'(x).$$
5. **Derivative of Inverse Function:** If f is a bijective differentiable function with $f'(x_0) \neq 0$:
$$(f^{-1})'(y_0) = \frac{1}{f'(x_0)} = \frac{1}{f'(f^{-1}(y_0))} \quad \text{where } y_0 = f(x_0).$$

5.4 Fundamental Theorems of Differential Calculus

Theorem · Fermat's Theorem on Local Extrema

Let $f : (a, b) \longrightarrow \mathbb{R}$ be differentiable on the open interval (a, b). If f attains a **local extremum** (local maximum or local minimum) at an interior point $c \in (a, b)$, then:

$$f'(c) = 0. \tag{5.5}$$

Points where $f'(c) = 0$ are called **critical points** (or stationary points).

Theorem · Rolle's Theorem (1691)

Let $f : [a, b] \longrightarrow \mathbb{R}$ be a function satisfying:

1. f is continuous on the closed interval $[a, b]$;
2. f is differentiable on the open interval (a, b);
3. $f(a) = f(b)$.

Then there exists at least one interior point $c \in (a, b)$ such that:

$$f'(c) = 0. \tag{5.6}$$

Theorem · The Mean Value Theorem (Théorème des Accroissements Finis / TAF)

Let $f : [a, b] \longrightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). Then there exists at least one point $c \in (a, b)$ such that:

$$f(b) - f(a) = f'(c) \cdot (b - a) \iff f'(c) = \frac{f(b) - f(a)}{b - a}. \tag{5.7}$$

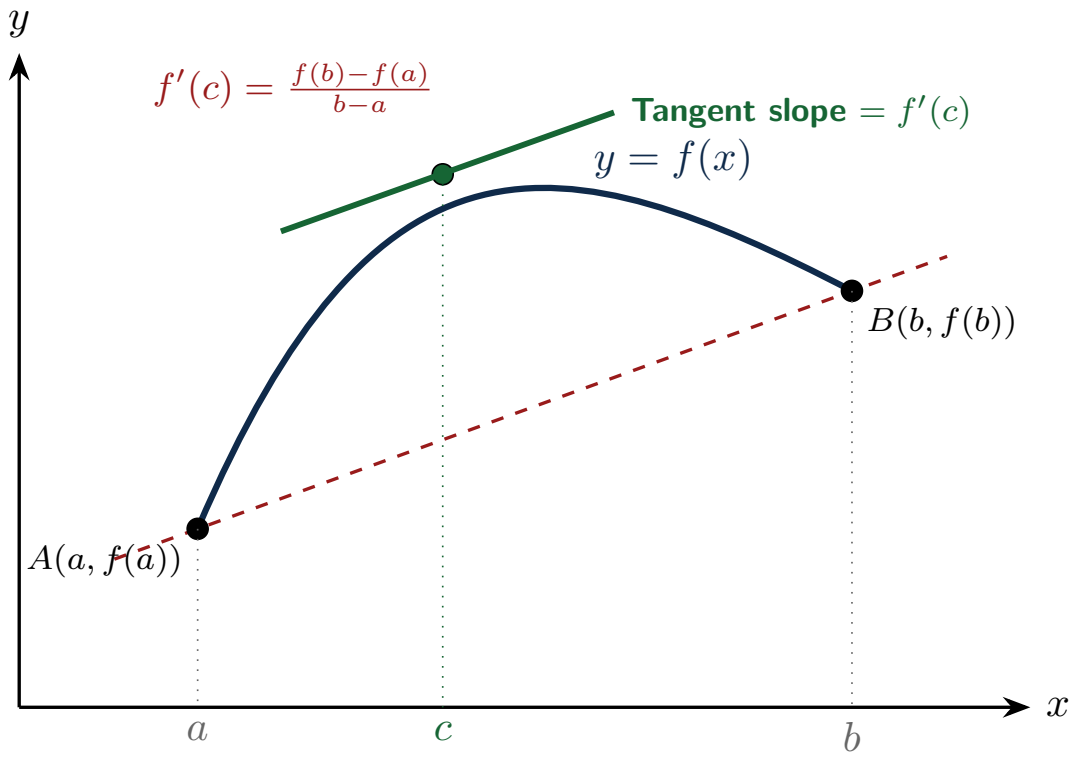


Figure 5.3: Rolle's and Mean Value Theorem Geometry: The secant chord connecting endpoint coordinates $(a, f(a))$ and $(b, f(b))$ has slope $\frac{f(b)-f(a)}{b-a}$. There exists at least one interior point $c \in (a, b)$ where the tangent line to the curve is strictly parallel to the secant line ($f'(c) =$ slope).

Proposition / Lemma · Corollaries of the Mean Value Theorem

Let f be differentiable on an interval I:

1. **Constant Function Criterion:** $\forall x \in I, f'(x) = 0 \iff f$ is constant on I.
2. **Monotonicity Criteria:**
   - $f'(x) \geq 0$ on $I \iff f$ is monotonically increasing on I.
   - $f'(x) \leq 0$ on $I \iff f$ is monotonically decreasing on I.
   - $f'(x) > 0$ on $I \implies f$ is strictly increasing on I.
3. **Mean Value Inequality (Inégalité des Accroissements Finis):** If $|f'(x)| \leq M$ for all $x \in I$, then for all $x, y \in I$:
$$|f(x) - f(y)| \leq M|x - y|.$$
(The function is Lipschitz continuous with Lipschitz constant M).

Theorem · Cauchy's Generalized Mean Value Theorem and L'Hôpital's Rule

1. **Cauchy's Extended Mean Value Theorem:** If $f, g : [a, b] \longrightarrow \mathbb{R}$ are continuous on $[a, b]$ and differentiable on (a, b), and $g'(x) \neq 0$ on (a, b), then there exists $c \in (a, b)$ such that:
$$\frac{f(b) - f(a)}{g(b) - g(a)} = \frac{f'(c)}{g'(c)}.$$
2. **L'Hôpital's Rule:** If $\lim_{x \to x_0} f(x) = \lim_{x \to x_0} g(x) = 0$ (or $\pm\infty$), and $\lim_{x \to x_0} \frac{f'(x)}{g'(x)} = L \in$

$\mathbb{R} \cup \{\pm\infty\}$, then:

$$\lim_{x \to x_0} \frac{f(x)}{g(x)} = \lim_{x \to x_0} \frac{f'(x)}{g'(x)} = L.$$

5.5 Worked Examples with Step-by-Step Proofs

Worked Example 5.1 · Rigorous Proof of Rolle's Theorem from Weierstrass's Theorem

Provide the complete, rigorous analytical proof of Rolle's Theorem using the Extreme Value Theorem and Fermat's Theorem.

Solution and Proof. Let $f : [a, b] \longrightarrow \mathbb{R}$ satisfy the three hypotheses of Rolle's Theorem: (1) f is continuous on $[a, b]$, (2) f is differentiable on (a, b), and (3) $f(a) = f(b)$.
By the **Weierstrass Extreme Value Theorem**, because f is continuous on the compact interval $[a, b]$, it is bounded and attains both its absolute minimum m and absolute maximum M at points $x_{\min}, x_{\max} \in [a, b]$:

$$m = f(x_{\min}) = \min_{x \in [a,b]} f(x), \qquad M = f(x_{\max}) = \max_{x \in [a,b]} f(x).$$

We now examine two mutually exclusive and exhaustive cases:

- **Case 1: $m = M$ (Constant Function).** If the minimum equals the maximum, then for all $x \in [a, b]$, $m \leq f(x) \leq M \implies f(x) = m$. Thus f is a constant function on $[a, b]$. Its derivative is identically zero everywhere on (a, b):

  $$\forall x \in (a, b), \quad f'(x) = 0.$$

  Any point $c \in (a, b)$ satisfies $f'(c) = 0$.

- **Case 2: $m < M$ (Non-constant Function).** Since $f(a) = f(b)$, it is impossible for both extrema to occur simultaneously at the boundary endpoints $\{a, b\}$. Indeed, if both occurred at the endpoints, we would have $m = f(a) = f(b) = M$, which contradicts $m < M$. Therefore, at least one of the two extrema must occur at an **interior point** $c \in (a, b)$:

  – Either $x_{\max} \in (a, b)$, in which case we set $c = x_{\max}$;
  – Or $x_{\min} \in (a, b)$, in which case we set $c = x_{\min}$.

  Since $c \in (a, b)$, f has a local extremum at the interior point c. Moreover, by hypothesis (2), f is differentiable at c. By **Fermat's Theorem on Local Extrema**, the derivative at an interior extremum must vanish:

  $$f'(c) = 0.$$

In both cases, we have established the existence of at least one point $c \in (a, b)$ such that $\mathbf{f'(c) = 0}$.
■

Worked Example 5.2 · Rigorous Proof of the Mean Value Theorem from Rolle's Theorem

Derive the Mean Value Theorem by constructing an auxiliary function that satisfies Rolle's Theorem.

Solution and Proof. Let $f : [a, b] \longrightarrow \mathbb{R}$ be continuous on $[a, b]$ and differentiable on (a, b). The secant line connecting $(a, f(a))$ and $(b, f(b))$ has equation:

$$y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$$

Consider the auxiliary function $\phi : [a, b] \longrightarrow \mathbb{R}$ measuring the vertical deviation between the curve $y = f(x)$ and the secant line:

$$\phi(x) = f(x) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(x - a)\right].$$

Let us verify the three hypotheses of Rolle's Theorem for ϕ:

1. **Continuity on $[a, b]$:** ϕ is the difference of f (continuous on $[a, b]$) and an affine linear function (continuous on $\mathbb{R}$). Hence ϕ is continuous on $[a, b]$.
2. **Differentiability on (a, b):** ϕ is differentiable on (a, b) as the difference of differentiable functions, with derivative:
$$\phi'(x) = f'(x) - \frac{f(b) - f(a)}{b - a}.$$
3. **Equal Endpoint Values:**
$$\phi(a) = f(a) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(a - a)\right] = f(a) - f(a) = 0.$$
$$\phi(b) = f(b) - \left[f(a) + \frac{f(b) - f(a)}{b - a}(b - a)\right] = f(b) - [f(a) + f(b) - f(a)] = 0.$$
Thus $\phi(a) = \phi(b) = 0$.

All three conditions of Rolle's Theorem are satisfied by ϕ. Therefore, there exists at least one point $c \in (a, b)$ such that:

$$\phi'(c) = 0 \implies f'(c) - \frac{f(b) - f(a)}{b - a} = 0 \implies \mathbf{f'(c) = \frac{f(b) - f(a)}{b - a}}. \quad \blacksquare$$

Worked Example 5.3 · Proving Classical Inequalities via the Mean Value Theorem

Using the Mean Value Theorem:

1. Prove that for all real numbers $x > 0$:
$$\frac{x}{1 + x} < \ln(1 + x) < x.$$
2. Prove that for all $x \in \mathbb{R}$, $|\sin x| \leq |x|$.

Solution and Proof. **1. Logarithmic Inequality:** Fix $x > 0$ and consider the function $f(t) = \ln(1 + t)$ on the interval $[0, x]$.

- f is continuous on $[0, x]$ and differentiable on $(0, x)$, with derivative $f'(t) = \frac{1}{1+t}$.
- By the Mean Value Theorem, there exists $c \in (0, x)$ such that:
$$\frac{f(x) - f(0)}{x - 0} = f'(c) \implies \frac{\ln(1 + x) - \ln(1)}{x} = \frac{1}{1 + c} \implies \ln(1 + x) = \frac{x}{1 + c}.$$

- Since $0 < c < x$, we have $1 < 1 + c < 1 + x$. Inverting the inequalities:

$$\frac{1}{1+x} < \frac{1}{1+c} < 1.$$

Multiplying through by $x > 0$:

$$\frac{\mathbf{x}}{\mathbf{1}+\mathbf{x}} < \ln(\mathbf{1}+\mathbf{x}) < \mathbf{x}.$$

2. Sine Inequality: Consider $g(t) = \sin t$. For $x = 0$, $|\sin 0| = 0 \leq 0$ holds trivially. For $x \neq 0$, apply the Mean Value Theorem on the interval between 0 and x:

$$\frac{\sin x - \sin 0}{x - 0} = \cos(c) \quad \text{for some } c \text{ between } 0 \text{ and } x.$$

Taking absolute values:

$$\frac{|\sin x|}{|x|} = |\cos c|.$$

Since $|\cos c| \leq 1$ for all real c:

$$\frac{|\sin x|}{|x|} \leq 1 \implies |\sin \mathbf{x}| \leq |\mathbf{x}|. \quad \blacksquare$$

Worked Example 5.4 · Bijective Inverse Function Derivative

Consider the function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined by:

$$f(x) = x^3 + 2x - 1.$$

1. Prove that f is a bijection from $\mathbb{R}$ onto $\mathbb{R}$.
2. Calculate $f(1)$.
3. Compute the derivative of the inverse function at $y_0 = 2$, namely $(f^{-1})'(2)$.

Solution and Proof. **1. Proof of Bijectivity:** f is a polynomial, hence continuous and infinitely differentiable on $\mathbb{R}$. Its derivative is:

$$f'(x) = 3x^2 + 2.$$

For all $x \in \mathbb{R}$, $3x^2 \geq 0 \implies f'(x) \geq 2 > 0$. Since $f'(x) > 0$ everywhere, f is **strictly increasing** on $\mathbb{R}$. Furthermore, examining limits at infinity:

$$\lim_{x \to -\infty} f(x) = -\infty, \qquad \lim_{x \to +\infty} f(x) = +\infty.$$

By the **Continuous Bijection Theorem**, a strictly increasing continuous function on $\mathbb{R}$ is a bijection from $\mathbb{R}$ onto $\mathbb{R}$.

2. Evaluation at $x = 1$:

$$f(1) = 1^3 + 2(1) - 1 = 1 + 2 - 1 = \mathbf{2}.$$

3. Derivative of the Inverse Function at $y_0 = 2$: Since $f(1) = 2$, we have $f^{-1}(2) = 1$. Using the formula for the derivative of the inverse function:

$$(f^{-1})'(2) = \frac{1}{f'(f^{-1}(2))} = \frac{1}{f'(1)}.$$

Compute $f'(1)$:

$$f'(1) = 3(1)^2 + 2 = 3 + 2 = 5 \neq 0.$$

Therefore:

$$(\mathbf{f^{-1}})'(\mathbf{2}) = \frac{\mathbf{1}}{\mathbf{5}} = \mathbf{0.2}. \quad \blacksquare$$

Chapter 6

Taylor Expansions, Asymptotic Developments & Riemann Integration

6.1 Higher-Order Derivatives and Leibniz's Formula

Let $f : I \longrightarrow \mathbb{R}$ be an n-times differentiable function on an interval I. The successive derivatives are defined recursively by $f^{(0)} = f$, $f^{(1)} = f'$, and $f^{(k+1)} = (f^{(k)})'$. If $f^{(n)}$ is continuous on I, we write $f \in \mathcal{C}^n(I)$. If f is infinitely differentiable, $f \in \mathcal{C}^\infty(I)$.



(a) **Isaac Newton (1642–1727)**



(b) **Gottfried Leibniz (1646–1716)**



(c) **Brook Taylor (1685–1731)**

Figure 6.1: Architects of the Infinitesimal Calculus: Sir Isaac Newton (fluxions and physics foundations), Gottfried Wilhelm Leibniz (differential and integral notation $\mathrm{d}x, \int$), and Brook Taylor (originator of Taylor polynomial expansions in 1715).

Theorem · Leibniz's Formula for Product Differentiation

Let $u, v : I \longrightarrow \mathbb{R}$ be two functions of class $\mathcal{C}^n(I)$. The n-th derivative of their product $u \cdot v$ is given by the binomial convolution:

$$(u \cdot v)^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} u^{(k)}(x) \cdot v^{(n-k)}(x), \qquad \text{where } \binom{n}{k} = \frac{n!}{k!(n-k)!}. \tag{6.1}$$

6.2 Taylor's Theorems: Lagrange and Peano (Young) Remainders

Theorem · Taylor's Theorem with Lagrange Remainder

Let $f \in \mathcal{C}^n([a, b])$ such that $f^{(n+1)}$ exists on the open interval (a, b). For any $x_0, x \in [a, b]$, there exists at least one real number c strictly between x_0 and x such that:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}. \tag{6.2}$$

The polynomial $P_n(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k$ is the **n-th Taylor polynomial**, and $R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x - x_0)^{n+1}$ is the **Lagrange remainder**.

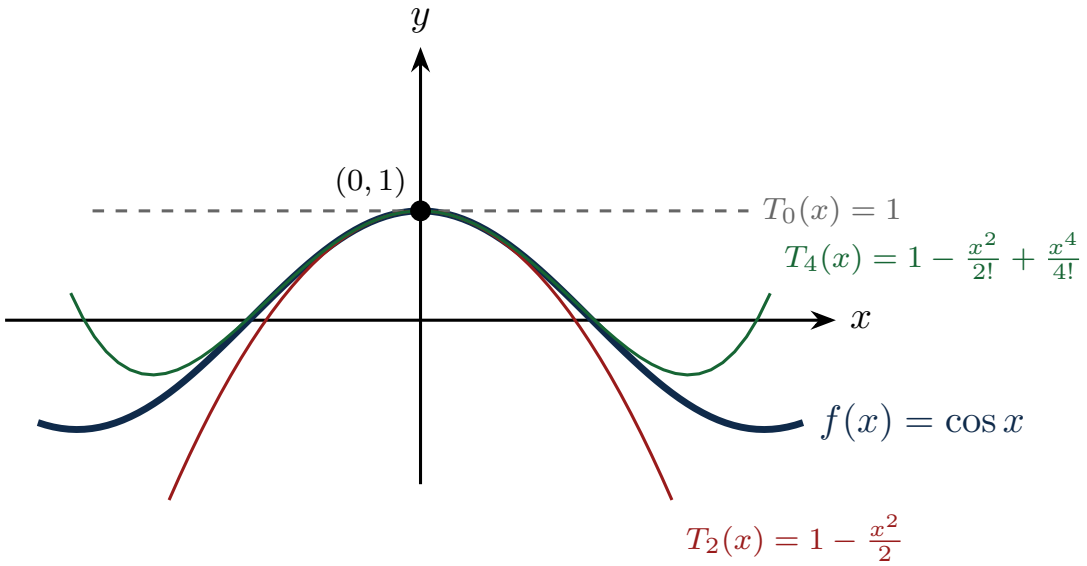


Figure 6.2: Taylor Polynomial Approximations of $\cos x$ at $x_0 = 0$: The zero-order polynomial $T_0(x) = 1$, second-order parabola $T_2(x) = 1 - \frac{x^2}{2}$, and fourth-order polynomial $T_4(x) = 1 - \frac{x^2}{2!} + \frac{x^4}{4!}$ progressively hug the true cosine curve over wider domains.

Theorem · Taylor-Young Formula (Local Expansion with Peano Remainder)

If f is n-times differentiable at x_0, then as $x \to x_0$:

$$f(x) = \sum_{k=0}^{n} \frac{f^{(k)}(x_0)}{k!}(x - x_0)^k + o((x - x_0)^n) \quad \text{as } x \to x_0, \tag{6.3}$$

where $o((x - x_0)^n) = (x - x_0)^n \varepsilon(x)$ with $\lim_{x \to x_0} \varepsilon(x) = 0$ (Landau's small-o notation).

6.3 Limited Developments (Développements Limités at 0)

Definition · Limited Development of Order n at 0 ($\mathrm{DL}_n(0)$)

A function f admits a **limited development** of order $n \in \mathbb{N}$ in the neighborhood of 0 if there exists a polynomial $P_n(x) = a_0 + a_1 x + a_2 x^2 + \cdots + a_n x^n$ of degree $\leq n$ such that:

$$f(x) = P_n(x) + x^n \varepsilon(x) \quad \text{with} \quad \lim_{x \to 0} \varepsilon(x) = 0. \tag{6.4}$$

The polynomial $P_n(x)$ is **strictly unique**. If f is n-times differentiable, $a_k = \frac{f^{(k)}(0)}{k!}$ (Maclaurin expansion).

Proposition / Lemma · Reference Maclaurin Expansions at $x = 0$

As $x \to 0$, the following standard limited developments hold:

$$e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots + \frac{x^n}{n!} + o(x^n) \tag{6.5}$$

$$\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots + (-1)^p \frac{x^{2p+1}}{(2p+1)!} + o(x^{2p+2}) \tag{6.6}$$

$$\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots + (-1)^p \frac{x^{2p}}{(2p)!} + o(x^{2p+1}) \tag{6.7}$$

$$\ln(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - \frac{x^4}{4} + \cdots + (-1)^{n-1} \frac{x^n}{n} + o(x^n) \tag{6.8}$$

$$(1+x)^\alpha = 1 + \alpha x + \frac{\alpha(\alpha-1)}{2!}x^2 + \cdots + \frac{\alpha(\alpha-1)\cdots(\alpha-n+1)}{n!}x^n + o(x^n) \tag{6.9}$$

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \cdots + x^n + o(x^n) \tag{6.10}$$

$$\frac{1}{1+x} = 1 - x + x^2 - x^3 + \cdots + (-1)^n x^n + o(x^n) \tag{6.11}$$

$$\arctan x = x - \frac{x^3}{3} + \frac{x^5}{5} - \cdots + (-1)^p \frac{x^{2p+1}}{2p+1} + o(x^{2p+2}) \tag{6.12}$$

6.4 The Riemann Integral and Fundamental Theorem of Calculus



(a) **Leonhard Euler (1707–1783)**



(b) **Bernhard Riemann (1826–1866)**

Figure 6.3: Pioneers of Integral Analysis: Leonhard Euler (master of analytical transformations and infinite sums) and Bernhard Riemann (creator of the modern definition of the definite integral via partition step sums).

Definition · The Riemann Integral

Let $f : [a,b] \longrightarrow \mathbb{R}$ be a bounded function. A **subdivision** (partition) of $[a,b]$ is a finite ordered sequence $\sigma = \{a = x_0 < x_1 < x_2 < \cdots < x_n = b\}$. For any choice of evaluation tags $\xi_i \in [x_{i-1}, x_i]$, the **Riemann sum** is:

$$S(f, \sigma, \xi) = \sum_{i=1}^{n} f(\xi_i) \cdot (x_i - x_{i-1}) = \sum_{i=1}^{n} f(\xi_i) \Delta x_i. \tag{6.13}$$

f is **Riemann integrable** on $[a, b]$ if the Riemann sums converge to a unique real limit as the mesh size $\max_i \Delta x_i \to 0$:

$$\int_a^b f(x)\,\mathrm{d}x = \lim_{\|\sigma\|\to 0} \sum_{i=1}^{n} f(\xi_i)\Delta x_i. \tag{6.14}$$

Integrability Criterion: Every continuous function on $[a, b]$ ($f \in \mathcal{C}^0([a, b])$) and every piecewise continuous monotonic function is Riemann integrable.

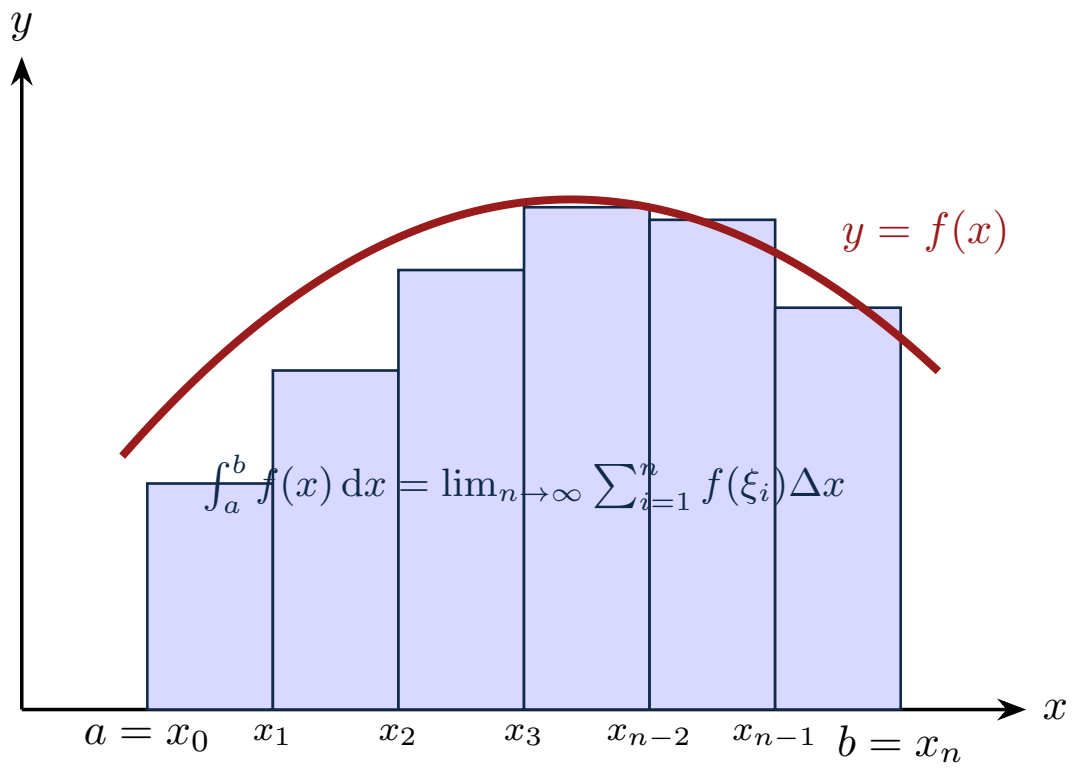


Figure 6.4: Riemann Integral as the Limit of Rectangular Approximations: Subdividing $[a, b]$ into n subintervals and summing the areas of rectangular slices converges to the exact geometric area under $y = f(x)$.

Theorem · The Fundamental Theorem of Calculus (FTC)

1. **First Part (Existence of Antiderivatives):** Let $f : [a, b] \longrightarrow \mathbb{R}$ be continuous. The function $F : [a, b] \longrightarrow \mathbb{R}$ defined by:

   $$F(x) = \int_a^x f(t)\,\mathrm{d}t$$

   is continuous on $[a, b]$, differentiable on (a, b), and its derivative is identically f:

   $$F'(x) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\int_a^x f(t)\,\mathrm{d}t\right) = f(x) \quad \text{for all } x \in [a, b].$$

2. **Second Part (Newton-Leibniz Formula):** If G is any antiderivative of f on $[a, b]$ (so $G'(x) = f(x)$):

   $$\int_a^b f(t)\,\mathrm{d}t = G(b) - G(a) = \big[G(t)\big]_a^b. \tag{6.15}$$

Proposition / Lemma · Essential Integration Techniques

1. **Integration by Parts (IPP):** Let $u, v \in \mathcal{C}^1([a,b])$:

$$\int_a^b u(t)v'(t)\,\mathrm{d}t = \big[u(t)v(t)\big]_a^b - \int_a^b u'(t)v(t)\,\mathrm{d}t. \tag{6.16}$$

2. **Integration by Substitution (Change of Variable):** Let $\phi : [\alpha,\beta] \longrightarrow [a,b]$ be a function of class $\mathcal{C}^1$ with $\phi(\alpha) = a$ and $\phi(\beta) = b$, and let $f \in \mathcal{C}^0([a,b])$:

$$\int_a^b f(x)\,\mathrm{d}x = \int_\alpha^\beta f(\phi(t)) \cdot \phi'(t)\,\mathrm{d}t. \tag{6.17}$$

6.5 Worked Examples with Step-by-Step Proofs

Worked Example 6.1 · High-Order Derivative Using Leibniz's Product Rule

Compute the n-th derivative of the function:

$$f(x) = x^2 e^{3x}.$$

Solution and Proof. Write $f(x) = u(x) \cdot v(x)$ where:

$$u(x) = x^2, \qquad v(x) = e^{3x}.$$

Compute the successive derivatives of $u(x)$:

$$u^{(0)}(x) = x^2, \quad u^{(1)}(x) = 2x, \quad u^{(2)}(x) = 2, \quad \text{and} \quad u^{(k)}(x) = 0 \quad \text{for all } k \geq 3.$$

Compute the successive derivatives of $v(x)$:

$$v^{(k)}(x) = 3^k e^{3x} \quad \text{for all } k \geq 0.$$

Applying **Leibniz's Formula**:

$$f^{(n)}(x) = \sum_{k=0}^{n} \binom{n}{k} u^{(k)}(x) v^{(n-k)}(x).$$

Since $u^{(k)}(x) = 0$ for all $k \geq 3$, only the first three terms ($k = 0, 1, 2$) survive:

$$\begin{aligned} f^{(n)}(x) &= \binom{n}{0} u(x)v^{(n)}(x) + \binom{n}{1} u'(x)v^{(n-1)}(x) + \binom{n}{2} u''(x)v^{(n-2)}(x) \\ &= 1 \cdot (x^2) \cdot \left(3^n e^{3x}\right) + n \cdot (2x) \cdot \left(3^{n-1} e^{3x}\right) + \frac{n(n-1)}{2} \cdot (2) \cdot \left(3^{n-2} e^{3x}\right). \end{aligned}$$

Factor out the common exponential and power term $3^{n-2}e^{3x}$:

$$\begin{aligned} f^{(n)}(x) &= 3^{n-2} e^{3x} \left[3^2 x^2 + 2n \cdot 3x + n(n-1)\right] \\ &= \mathbf{3^{n-2} e^{3x}} \left[\mathbf{9x^2 + 6nx + n(n-1)}\right]. \quad \blacksquare \end{aligned}$$

Worked Example 6.2 · Indeterminate Limit via Limited Development

Compute using limited developments the limit:

$$L = \lim_{x\to 0} \frac{\sin x - x\cos x}{x^3}.$$

Solution and Proof. As $x \to 0$, the numerator approaches $\sin 0 - 0 = 0$ and the denominator $x^3 \to 0$, producing an indeterminate form of type $\left[\frac{0}{0}\right]$. Since the denominator is of degree 3, we expand the numerator to order 3 ($\mathrm{DL}_3(0)$).
Recall the reference Maclaurin expansions to order 3:

$$\sin x = x - \frac{x^3}{6} + o(x^3),$$
$$\cos x = 1 - \frac{x^2}{2} + o(x^2).$$

Multiply $\cos x$ by x:

$$x\cos x = x\left(1 - \frac{x^2}{2} + o(x^2)\right) = x - \frac{x^3}{2} + o(x^3).$$

Subtract the two expressions:

$$\begin{aligned}\sin x - x\cos x &= \left(x - \frac{x^3}{6} + o(x^3)\right) - \left(x - \frac{x^3}{2} + o(x^3)\right)\\ &= (x-x) + \left(-\frac{1}{6} + \frac{1}{2}\right)x^3 + o(x^3)\\ &= \left(-\frac{1}{6} + \frac{3}{6}\right)x^3 + o(x^3) = \frac{2}{6}x^3 + o(x^3) = \frac{1}{3}x^3 + o(x^3).\end{aligned}$$

Now divide by the denominator x^3:

$$\frac{\sin x - x\cos x}{x^3} = \frac{\frac{1}{3}x^3 + o(x^3)}{x^3} = \frac{1}{3} + \frac{o(x^3)}{x^3} = \frac{1}{3} + \varepsilon(x),$$

where $\lim_{x\to 0}\varepsilon(x) = 0$. Taking the limit as $x \to 0$:

$$\mathbf{L = \lim_{x\to 0} \frac{\sin x - x\cos x}{x^3} = \frac{1}{3}}. \quad \blacksquare$$

Worked Example 6.3 · Integration by Parts: Wallis Integrals

Consider the sequence of **Wallis Integrals** defined for $n \in \mathbb{N}$ by:

$$W_n = \int_0^{\pi/2} \sin^n(x)\,\mathrm{d}x.$$

1. Compute W_0 and W_1.
2. Using integration by parts, establish the recurrence relation:

$$W_n = \frac{n-1}{n} W_{n-2} \quad \text{for all } n \geq 2.$$

Solution and Proof. 1. Direct Evaluation of W_0 and W_1:

$$W_0 = \int_0^{\pi/2} 1\,\mathrm{d}x = [x]_0^{\pi/2} = \frac{\pi}{2}.$$

$$W_1 = \int_0^{\pi/2} \sin x \, \mathrm{d}x = [-\cos x]_0^{\pi/2} = -\cos(\pi/2) - (-\cos 0) = 0 - (-1) = \mathbf{1}.$$

2. Integration by Parts for $n \geq 2$: Split the integrand: $\sin^n x = \sin^{n-1} x \cdot \sin x$. Set:

$$\begin{aligned} u(x) &= \sin^{n-1} x \implies u'(x) = (n-1)\sin^{n-2} x \cos x, \\ v'(x) &= \sin x \implies v(x) = -\cos x. \end{aligned}$$

Applying the integration by parts formula:

$$W_n = \big[-\sin^{n-1}(x)\cos(x) \big]_0^{\pi/2} - \int_0^{\pi/2} (n-1)\sin^{n-2}(x)\cos(x) \cdot (-\cos x) \, \mathrm{d}x.$$

Evaluate the boundary bracket:

- At $x = \pi/2$: $\cos(\pi/2) = 0$.
- At $x = 0$: for $n \geq 2$, $n - 1 \geq 1$, so $\sin^{n-1}(0) = 0$.

Thus the boundary term vanishes identically: $\big[-\sin^{n-1}(x)\cos(x) \big]_0^{\pi/2} = 0$. The integral term becomes:

$$W_n = (n-1) \int_0^{\pi/2} \sin^{n-2}(x)\cos^2(x) \, \mathrm{d}x.$$

Substitute the trigonometric identity $\cos^2 x = 1 - \sin^2 x$:

$$\begin{aligned} W_n &= (n-1) \int_0^{\pi/2} \sin^{n-2}(x)(1 - \sin^2 x) \, \mathrm{d}x \\ &= (n-1) \int_0^{\pi/2} \sin^{n-2}(x) \, \mathrm{d}x - (n-1) \int_0^{\pi/2} \sin^n(x) \, \mathrm{d}x \\ &= (n-1)W_{n-2} - (n-1)W_n. \end{aligned}$$

Rearranging terms in W_n:

$$W_n + (n-1)W_n = (n-1)W_{n-2} \implies nW_n = (n-1)W_{n-2}.$$

Dividing across by $n > 0$:

$$\mathbf{W_n = \frac{n-1}{n} W_{n-2}.} \quad \blacksquare$$

Worked Example 6.4 · Integration by Rational Substitution

Compute the definite integral:

$$I = \int_0^1 \frac{\mathrm{d}x}{1 + \sqrt{x}}.$$

Solution and Proof. The presence of the square root radical $\sqrt{x}$ suggests the change of variable:

$$u = \sqrt{x} \iff x = u^2.$$

Differentiating both sides:

$$\mathrm{d}x = 2u \, \mathrm{d}u.$$

Determine the new integration bounds:

- When $x = 0$: $u = \sqrt{0} = 0$.
- When $x = 1$: $u = \sqrt{1} = 1$.

Substitute into the integral:

$$I = \int_0^1 \frac{2u\,\mathrm{d}u}{1+u} = 2\int_0^1 \frac{u}{u+1}\,\mathrm{d}u.$$

Perform algebraic decomposition of the rational integrand by adding and subtracting 1:

$$\frac{u}{u+1} = \frac{(u+1)-1}{u+1} = 1 - \frac{1}{u+1}.$$

Now integrate term by term:

$$\begin{aligned} I &= 2\int_0^1 \left(1 - \frac{1}{u+1}\right)\mathrm{d}u = 2\Big[u - \ln(u+1)\Big]_0^1 \\ &= 2\Big[(1-\ln 2) - (0 - \ln 1)\Big] \\ &= 2(1-\ln 2) = \mathbf{2 - 2\ln 2 \approx 0.6137}. \quad \blacksquare \end{aligned}$$

Chapter 7

Mathematical Reference Tables and Formula Manual

7.1 Master Table of Standard Derivatives

Function $f(x)$	**Derivative $f'(x)$**	**Domain of Differentiability**
Constant: c	0	$\mathbb{R}$
Power: x^n $(n \in \mathbb{Z})$	nx^{n-1}	$\mathbb{R}$ (or $\mathbb{R}^*$ if $n < 0$)
Power: x^α $(\alpha \in \mathbb{R})$	$\alpha x^{\alpha-1}$	$(0, +\infty)$
Square root: $\sqrt{x}$	$\dfrac{1}{2\sqrt{x}}$	$(0, +\infty)$
Natural exponential: e^x	e^x	$\mathbb{R}$
General exponential: a^x $(a > 0)$	$a^x \ln(a)$	$\mathbb{R}$
Natural logarithm: $\ln \lvert x \rvert$	$\dfrac{1}{x}$	$\mathbb{R}^*$
Sine: $\sin(x)$	$\cos(x)$	$\mathbb{R}$
Cosine: $\cos(x)$	$-\sin(x)$	$\mathbb{R}$
Tangent: $\tan(x)$	$1 + \tan^2(x) = \dfrac{1}{\cos^2(x)}$	$x \neq \dfrac{\pi}{2} + k\pi$
Arcsine: $\arcsin(x)$	$\dfrac{1}{\sqrt{1-x^2}}$	$(-1, 1)$
Arccosine: $\arccos(x)$	$-\dfrac{1}{\sqrt{1-x^2}}$	$(-1, 1)$
Arctangent: $\arctan(x)$	$\dfrac{1}{1+x^2}$	$\mathbb{R}$
Hyperbolic sine: $\sinh(x)$	$\cosh(x)$	$\mathbb{R}$
Hyperbolic cosine: $\cosh(x)$	$\sinh(x)$	$\mathbb{R}$
Hyperbolic tangent: $\tanh(x)$	$1 - \tanh^2(x) = \dfrac{1}{\cosh^2(x)}$	$\mathbb{R}$

7.2 Master Table of Indefinite Integrals (Antiderivatives)

Integrand $f(x)$	**General Antiderivative $\int f(x)\,dx$**	**Validity Interval**
x^α $(\alpha \neq -1)$	$\dfrac{x^{\alpha+1}}{\alpha+1} + C$	$(0, +\infty)$
$\dfrac{1}{x}$	$\ln \lvert x \rvert + C$	$(-\infty, 0)$ or $(0, +\infty)$
$e^{\lambda x}$ $(\lambda \neq 0)$	$\dfrac{1}{\lambda} e^{\lambda x} + C$	$\mathbb{R}$
$\sin(x)$	$-\cos(x) + C$	$\mathbb{R}$

Integrand $f(x)$	**General Antiderivative $\int f(x)\,\mathrm{d}x$**	**Validity Interval**
$\cos(x)$	$\sin(x) + C$	$\mathbb{R}$
$\dfrac{1}{\cos^2(x)} = 1 + \tan^2(x)$	$\tan(x) + C$	$(-\pi/2, \pi/2) + k\pi$
$\dfrac{1}{1+x^2}$	$\arctan(x) + C$	$\mathbb{R}$
$\dfrac{1}{a^2+x^2}$ $(a > 0)$	$\dfrac{1}{a}\arctan\left(\dfrac{x}{a}\right) + C$	$\mathbb{R}$
$\dfrac{1}{\sqrt{1-x^2}}$	$\arcsin(x) + C$	$(-1, 1)$
$\dfrac{1}{\sqrt{a^2-x^2}}$ $(a > 0)$	$\arcsin\left(\dfrac{x}{a}\right) + C$	$(-a, a)$
$\dfrac{1}{\sqrt{x^2+1}}$	$\operatorname{argsinh}(x) + C = \ln\left(x + \sqrt{x^2+1}\right) + C$	$\mathbb{R}$
$\dfrac{1}{\sqrt{x^2-1}}$	$\operatorname{argcosh}(x) + C = \ln\left(x + \sqrt{x^2-1}\right) + C$	$(1, +\infty)$
$\dfrac{u'(x)}{u(x)}$	$\ln\lvert u(x)\rvert + C$	where $u(x) \neq 0$
$u'(x)[u(x)]^\alpha$ $(\alpha \neq -1)$	$\dfrac{[u(x)]^{\alpha+1}}{\alpha+1} + C$	where $u(x)$ valid
$u'(x)e^{u(x)}$	$e^{u(x)} + C$	$\mathbb{R}$

7.3 Master Table of Asymptotic Equivalents as $x \to 0$

Expression	**Equivalent as $x \to 0$**
$\sin(x)$	x
$\tan(x)$	x
$\arcsin(x)$	x
$\arctan(x)$	x
$\sinh(x)$	x
$\tanh(x)$	x
$1 - \cos(x)$	$\dfrac{x^2}{2}$
$\cosh(x) - 1$	$\dfrac{x^2}{2}$
$e^x - 1$	x
$\ln(1+x)$	x
$(1+x)^\alpha - 1$	αx
$\sqrt{1+x} - 1$	$\dfrac{x}{2}$